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Sulfur, silicon, and phosphorus in organic chemistry

Connections

➡ Building on

- Carbonyl chemistry [ch6](#), [ch10](#), & [ch11](#)
- Wittig reaction [ch11](#)
- Kinetic and thermodynamic control [ch12](#)
- Stereochemistry [ch14](#)
- Elimination reactions [ch17](#)
- Conjugate addition [ch22](#)
- Reduction [ch23](#)
- Chemistry of enol(ate)s [ch25](#) & [ch26](#)

Arriving at

- Organic S and Si chemistry
- S, Si, and P in the synthesis of alkenes
- Why *E/Z* control matters
- Ways to control *E/Z* geometry
- Equilibration of alkenes gives *trans*
- Effects of light and how we see
- Julia olefination and the Wittig reaction at work
- Reliable reduction of alkynes

➡ Looking forward to

- Diastereoselectivity [ch33](#)
- Pericyclic reactions [ch34](#) & [ch35](#)
- Fragmentations [ch36](#)
- Radicals and carbenes [ch37](#) & [ch38](#)
- Asymmetric synthesis [ch41](#)

Useful main group elements

Electronegativities

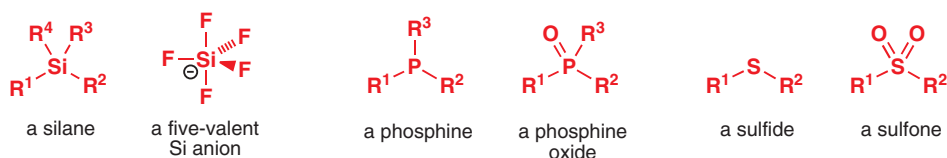
C (2.5)	N (3.0)	O (3.5)	F (4.0)
Si (1.8)	P (2.1)	S (2.5)	Cl (3.0)

Organic chemists make use of most elements in the periodic table: you have already seen organic compounds of Li, B, F, Na, Mg, Al, Si, P, S, Cl, K, Cu, Br, and I—but that's only the start. Three of the most important of these are sulfur, phosphorus, and silicon. They all form stable organic compounds and play nearly as important roles in organic chemistry as oxygen, nitrogen, and the halogens. They are second row elements, coming immediately below carbon, nitrogen, and oxygen, to which they have some similarity. Electronegativity (shown in the margin) diminishes from right to left and downwards.

The main difference from C, N, and O is that Si, P, and S can form more bonds than the first row elements. This is because they have more orbitals: the five 3d orbitals added to the 3s and three 3p orbitals. Silicon forms tetrahedral silanes, rather like alkanes, but also forms stable five-valent anions. Phosphorus forms phosphines, rather like amines, but also tetrahedral phosphine oxides. Sulfur can have any coordination number from zero to seven, forming sulfides, like ethers, and tetrahedral sulfones with six bonds to sulfur. And it is with sulfur that we start.

Spelling sulfur

If you look in the Oxford English dictionary you will see 'sulphur' (along with 'sulphuric', 'sulphate'...). These are peculiarly British spellings, and it was agreed some years ago that chemists the world over should use a uniform spelling: 'sulfur'.



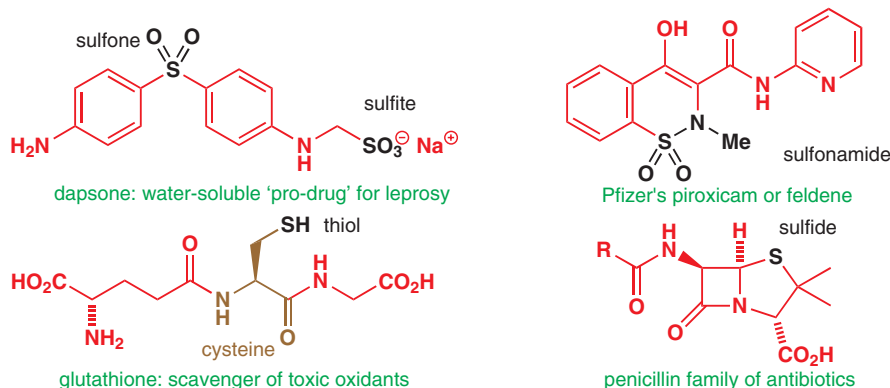
Sulfur: an element of contradictions

The first organosulfur compounds in this book were the dreadful smell of the skunk and the wonderful smell of the truffle, which pigs can detect through a metre of soil and which is so

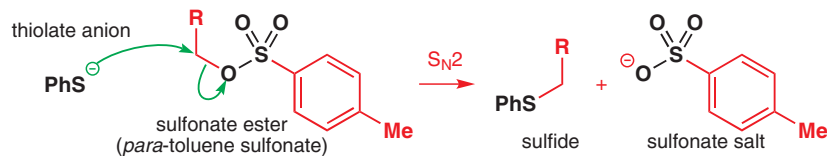
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

delightful that truffles cost more than their weight in gold. Sulfur compounds can be reducing or oxidizing agents, anions or cations, nucleophiles or electrophiles as well as foul- or sweet-smelling.

Useful sulfur compounds include the leprosy drug dapsone (Chapter 6), the arthritis drug feldene (Chapter 20), glutathione (Chapter 22), a scavenger of oxidizing agents that protects most living things against oxidation and contains the natural amino acid cysteine, and, of course, the famous antibiotics, the penicillins, mentioned in several chapters.



Important reactions include sulfur as nucleophile and leaving group in the $\text{S}_{\text{N}}2$ reaction, sulfonation of aromatic rings (Chapter 21), and formation and reduction of thioacetals (Chapter 23). This $\text{S}_{\text{N}}2$ reaction uses a sulfur nucleophile and a sulfur-based leaving group.



Some facts about sulfur

Sulfur is a p-block element in group VI (or 16 if you prefer) immediately below oxygen and between phosphorus and chlorine. It is natural for us to compare sulfur with oxygen but we will, strangely, compare it with carbon as well.

Sulfur is much less electronegative than oxygen; in fact, it has the same electronegativity as carbon, so it is no good trying to use the polarization of the C-S bond to explain anything! It forms reasonably strong bonds to carbon—strong enough for the compounds to be stable but weak enough for selective cleavage in the presence of the much stronger C-O bonds. It also forms fairly strong bonds to itself. Elemental crystalline yellow sulfur consists of S_8 molecules—eight-membered rings of sulfur atoms.

Because sulfur is in the second row of the periodic table it forms many types of compounds not available to oxygen. Compounds with S-S and S-halogen bonds are quite stable and can be isolated, unlike the unstable and often explosive O-halogen and O-O compounds. Sulfur's d orbitals allow it to have oxidation states of 0, 2, 4, or 6 and coordination numbers from 0 to 7. Here is a selection of compounds.

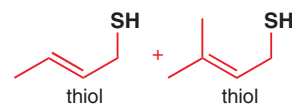
Compounds of sulfur

Oxidation state	S(II)			S(IV)		S(VI)		
coordination number	0	1	2	3	4	4	6	7
example	S^{2-}	RS^-	R_2S	$\text{R}_2\text{S=O}$	SF_4	R_2SO_2	SF_6	SF_7^-

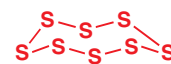
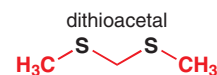
Sulfur is a very versatile element

As well as this variety of oxidation states, sulfur shows a sometimes surprising versatility in function. Simple S(II) compounds are good nucleophiles, as you would expect from the

the dreadful smell of the skunk



the delightful smell of the truffle

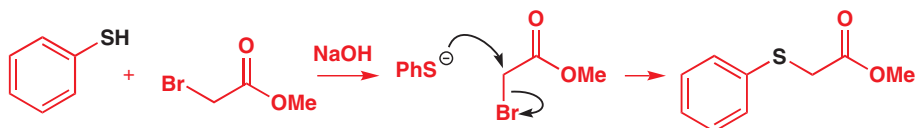


crystalline sulfur

Typical bond strengths, kJ mol^{-1}

X =	C	H	F	S
C-X	376	418	452	362
S-X	362	349	384	301

high-energy non-bonding lone pairs ($3sp^3$ rather than the $2sp^3$ of oxygen). A mixture of a thiol (RSH, the sulfur equivalent of an alcohol) and NaOH reacts with an alkyl halide to give the sulfide alone by nucleophilic attack of RS^- .



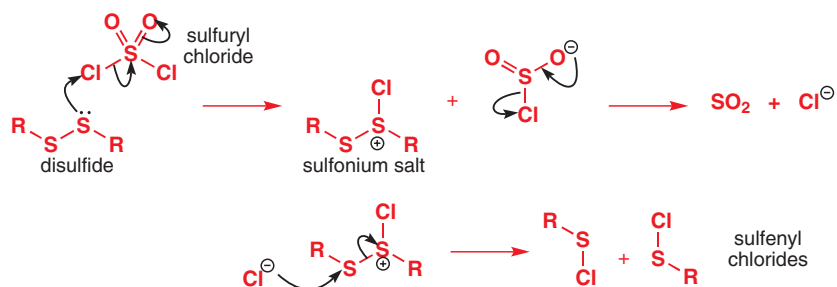
Thiols (RSH) are more acidic than alcohols so the first step is a rapid proton exchange between the thiol and hydroxide ion. The thiolate anion then carries out a very efficient S_N2 displacement on the alkyl bromide to give the sulfide. Notice that the thiolate anion does not attack the carbonyl group. Small basic oxyanions have high charge density and low-energy filled orbitals—they are hard nucleophiles that prefer to attack protons and carbonyl groups. Large, less basic thiolate anions have high-energy filled orbitals and are soft nucleophiles. They prefer to attack saturated carbon atoms. Thiols and thiolates are good soft nucleophiles.

➡ This 'hard or soft' nature of nucleophiles is discussed with S_N2 reactions in general in Chapter 15.

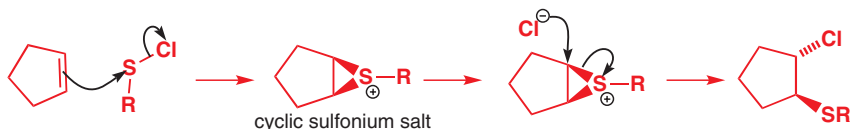
● Thiols (RSH) are more acidic than alcohols (ROH) but sulfur compounds are better nucleophiles than oxygen compounds towards saturated carbon atoms (S_N2).

■ Thionyl chloride $SOCl_2$ is electrophilic at sulfur but sulfuryl chloride SO_2Cl_2 is electrophilic at chlorine. Contradictory again!

They are also good soft electrophiles. Sulfenyl chlorides ($RSnCl$) are easily made from disulfides ($RS-SR$) and sulfuryl chloride (SO_2Cl_2). This S(VI) chloride has electrophilic chlorine atoms and is attacked by the nucleophilic disulfide to give two molecules of $RSnCl$ and gaseous SO_2 . There's a lot of sulfur chemistry here! We start with a nucleophilic attack by one sulfur atom of the disulfide. The intermediate contains a tricoordinate sulfur cation or sulfonium salt. The chloride ion now attacks the other sulfur atom of this intermediate and two molecules of $RSnCl$ result. Each atom of the original disulfide has formed an S-Cl bond. One sulfur atom was a nucleophile towards chlorine and the other an electrophile.

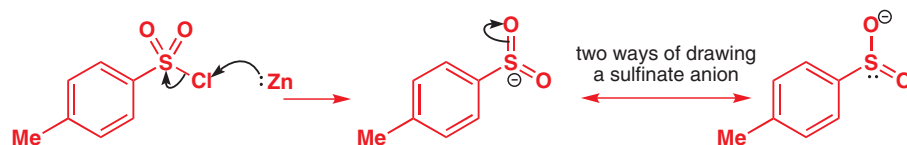


The product of this reaction, the sulfenyl chloride, is also a good soft electrophile towards carbon atoms, particularly towards alkenes. The reaction is very like bromination, with a three-membered cyclic sulfonium ion intermediate replacing the bromonium ion of Chapter 19. The reaction is stereospecific and the product is *anti*.

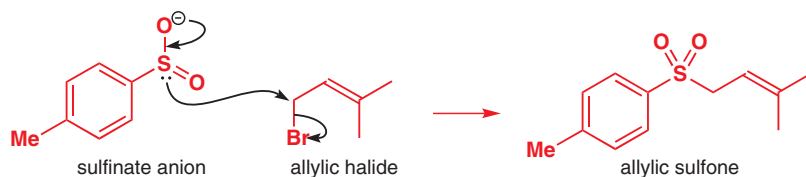


At higher oxidation states the compounds become harder electrophiles as the positive charge on sulfur increases. We have already mentioned tosyl (*para*-toluenesulfonyl) chloride, $TsCl$, as an electrophile for alkoxide ions in this chapter and in earlier chapters.

At this higher oxidation state it might seem unlikely that sulfur could also be a good nucleophile, but consider the result of reacting $TsCl$ with zinc metal. Zinc provides two electrons and turns the compound into an anion. This anion can also be drawn in two ways.



Surprisingly, this anion is also a good soft nucleophile and attacks saturated carbon atoms through the sulfur atom. In this case attack occurs at the less substituted end of an allylic bromide to give an allylic sulfone, which we will use later on.



- Sulfur compounds may be good nucleophiles and good electrophiles.

Sulfur-based functional groups

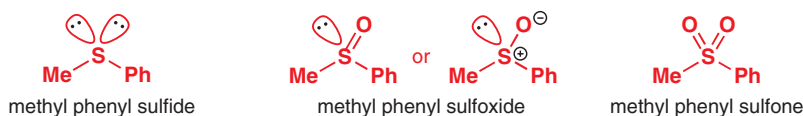
You have already met a number of sulfur-containing functional groups: the following list brings them together for reference.

Name	Structure	Importance	Example	Example details
thiol (or mercaptan)	RSH	strong smell, usually bad, but heavenly in low concentrations		smell and taste of coffee and grapefruit
thiolate anion	RS ⁻	good soft nucleophiles		
disulfide	RS-SR	cross-links proteins		cystine
sulfenyl chloride	RS-Cl	good soft electrophiles		
sulfide (or thioether)	R-S-R	molecular link		smell and taste of pineapple
sulfonium salt	R ₃ S ⁺	important reagents		ylid used in epoxidations
sulfoxide	R ₂ S=O or R ₂ S ⁺ -O ⁻	many reactions; can be chiral		DMSO (dimethyl sulfoxide)
sulfone	R ₂ SO ₂	anion-stabilizing group		
sulfonic acid	RSO ₂ OH	strong acids		p-toluenesulfonic acid, TsOH
sulfonyl chloride	RSO ₂ Cl	turns alcohols into leaving groups		p-toluenesulfonyl chloride, TsCl

As this chapter develops you will see other examples of the versatility of sulfur. You will see that it can be removed from organic compounds in either an oxidative or a reductive fashion, and you will see that it can stabilize anions or cations on adjacent carbon atoms. The stabilization of anions is the first main section of the chapter.

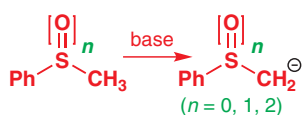
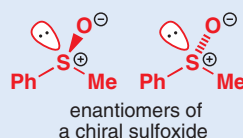
Sulfur-stabilized anions

The stabilization of anions by sulfides, sulfoxides, and sulfones is a theme that runs right through this chapter. Sulfur has six electrons in its outer shell. As a sulfide, therefore, the sulfur atom carries two lone pairs. In a sulfoxide, one of these lone pairs is used in a bond to an oxygen atom—sulfoxides can be represented in at least two alternative but equivalent ways. The sulfur atom in a sulfone uses both of its lone pairs in bonding to oxygen, and is usually represented with two S=O double bonds.

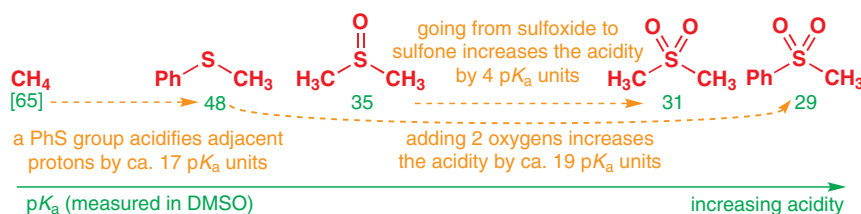


Chiral sulfoxides

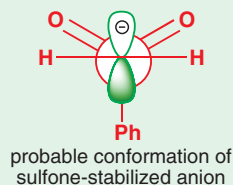
Sulfoxides have the potential for chirality—the tetrahedral sulfur atom is surrounded by four different groups (here Ph, Me, O, and the lone pair) and (unlike, say, the tetrahedral nitrogen atom of an amide) has a stable tetrahedral configuration. We will revisit chirality in sulfoxides later in the chapter.



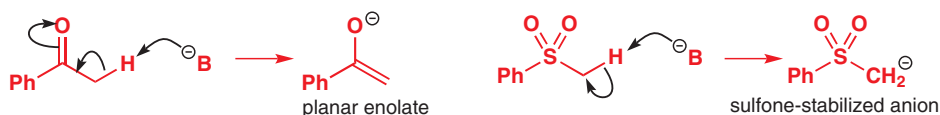
Treatment of any of these compounds with strong base produces an anion on what was the methyl group. How does the sulfur stabilize the anion? This question has been the subject of many debates and we have not got space to go into the details of all of them. There are at least two factors involved, and the first is evident from this chart of pK_a values for protons next to sulfone, sulfoxide, and sulfide functional groups.



■ Carbanions next to sulfones are planar, while anions next to sulfoxides and sulfides are thought to be pyramidal (sp^3 hybridized).



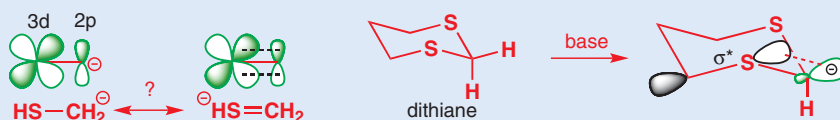
Clearly, the oxygen atoms are important—the best anion stabilizer is the sulfone, followed by the sulfoxide and then the sulfide. You could compare deprotonation of a sulfone with deprotonation of a ketone to give an enolate (Chapter 20). Enolates have a planar carbon atom and the anion is mainly on the oxygen atom. Sulfone-stabilized carbanions have two oxygen atoms and the anionic carbon atom is probably planar, with the negative charge in a p orbital aligned midway between the S=O bonds.



Yet the attached oxygen atoms cannot be the sole reason for the stability of anions next to sulfur because even the sulfide functional group also acidifies an adjacent proton quite significantly. There is some controversy over exactly why this should be, but the usual explanation is that polarization of the sulfur's 3s and 3p electrons (which are more diffuse, and therefore more polarizable, than the 2s and 2p electrons of oxygen) contributes to the stabilization.

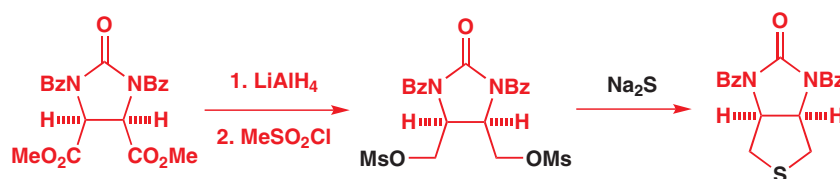
Anion stabilization by adjacent sulfur

It was long thought that delocalization into sulfur's empty 3d orbitals provided the anion stabilization required, but theoretical work in the last 20 years or so suggests this may not be the case. Thus, *ab initio* calculations suggest that the C–S bond in $-\text{CH}_2\text{SH}$ is longer than that in CH_3SH . The converse would be true if delocalization into the sulfur's d orbitals were important. Delocalization would shorten the bond because it would have partial double-bond character. More likely as an additional factor is delocalization into the σ^* orbital of the C–S bond on the other side of the sulfur atom—the equatorial proton of dithiane (see p. 662 for more on dithiane) is more acidic than the axial one, and the equatorial anion is more stable because it is delocalized into the C–S bond's σ^* orbital.



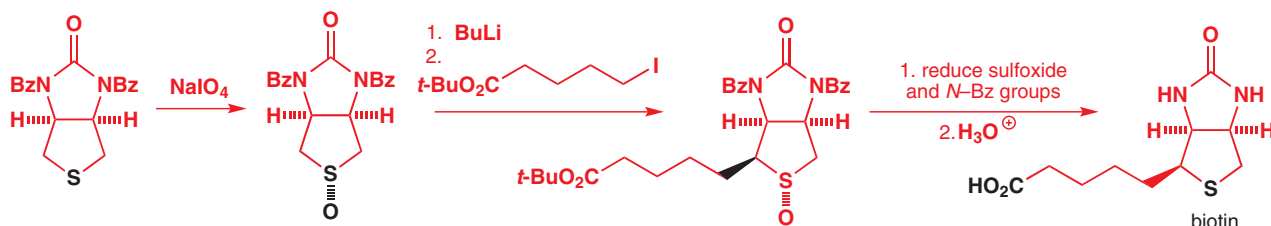
A sulfoxide-stabilized anion in a synthesis

A sulfoxide alkylation formed the key step of a synthesis of the important vitamin biotin. Biotin contains a five-membered heterocyclic sulfide fused to a second five-membered ring, and the bicyclic skeleton was easy to make from a simple symmetrical ester. The vital step is a double $\text{S}_{\text{N}}2$ reaction on primary carbon atoms.



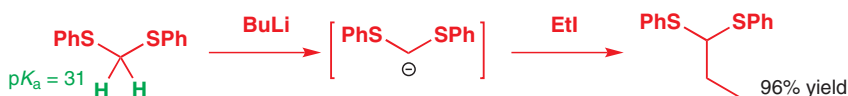
The next step was to introduce the alkyl chain—this was best done by first oxidizing the sulfide to a sulfoxide, using sodium periodate. The sulfoxide was then deprotonated with *n*-BuLi and alkylated with an alkyl iodide containing a carboxylic acid protected as its *tert*-butyl ester. Reduction of the sulfoxide and hydrolysis back to the free acid gave biotin.

➡ This synthesis involves some stereochemistry, which will be revisited in Chapter 32.

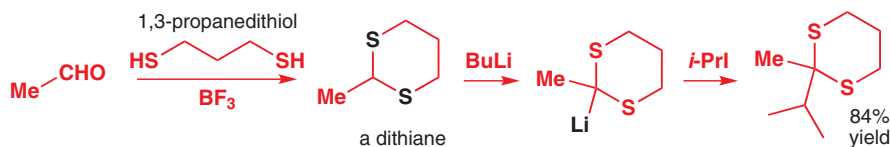


Thioacetals

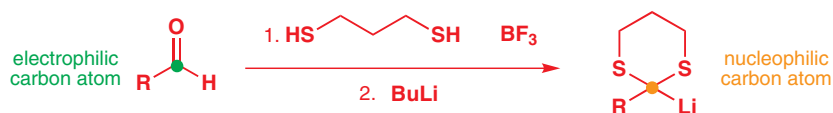
Although sulfide deprotonations are possible, the protons adjacent to *two* sulfide sulfur atoms are rather more acidic and alkylation of thioacetals is straightforward.



In general, thioacetals can be made in a similar way to ‘normal’ (oxygen-based) acetals—by treatment of an aldehyde or a ketone with a thiol and an acid catalyst—although a Lewis acid such as BF_3 is usually needed rather than a protic acid. The most easily made, most stable toward hydrolysis, and most reactive towards alkylation are cyclic thioacetals derived from 1,3-propanedithiol, known as dithianes.

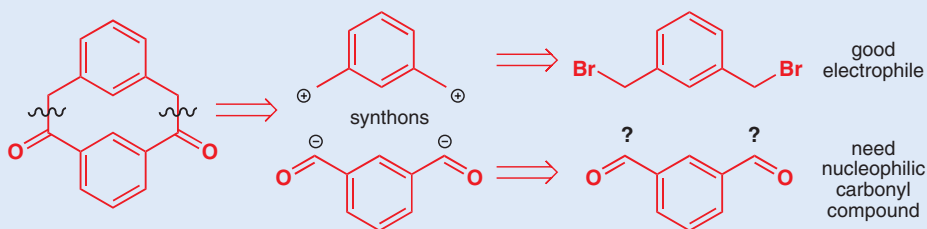


Dithianes are extremely important compounds in organic synthesis because *going from carbonyl compound to thioacetal inverts the polarity at the functionalized carbon atom*. Aldehydes, as you are well aware, are electrophiles at the $\text{C}=\text{O}$ carbon atom, but dithioacetals, through deprotonation to an anion, are nucleophilic at this same atom.

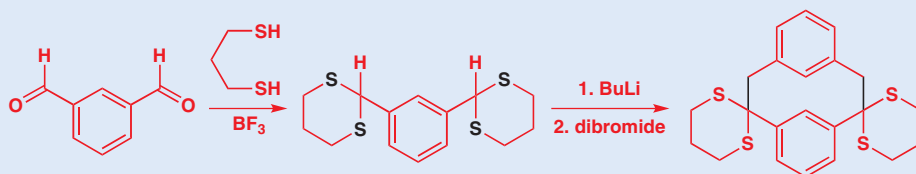


Dithianes in synthesis

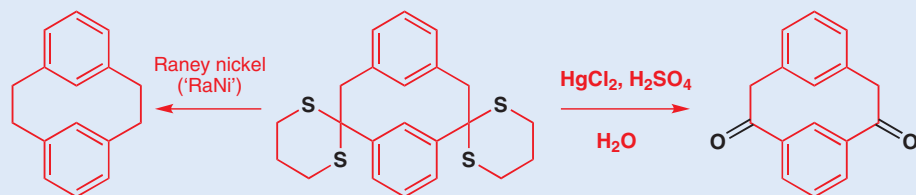
An example: chemists wanted to make this compound (a ‘metacyclophane’) because they wanted to study the independent rotation of the two benzene rings, which is hindered in such a small ring. An ideal way would be to join electrophilic benzylic bromides to nucleophilic carbonyl groups, if that were possible.



The dibromide and dialdehyde were both available—what they really wanted was a nucleophilic equivalent of the dialdehyde to react with the dibromide. So they made the dithiane. Sulfur is less basic than oxygen, so the protonated species is lower in concentration at a given pH, and the sulfur 3p lone pairs are less able to form a stable π bond to carbon than are the oxygen 2p lone pairs.



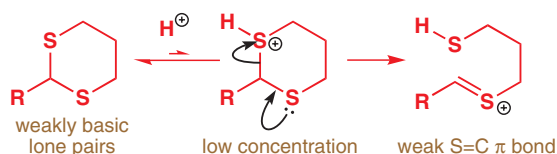
After the dithianes have been alkylated, they can be hydrolysed to give back the carbonyl groups. Alternatively, hydrogenation using Raney nickel replaces the thioacetal with a CH_2 group and gives the unsubstituted cyclophane.



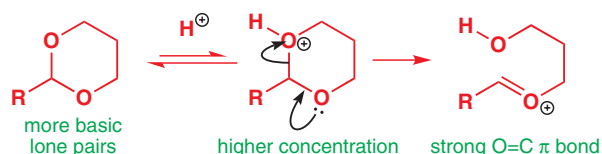
➡ See Chapter 28 for an explanation of these open ‘retrosynthetic’ arrows and the word ‘synthon’.

Dithianes are rather more stable than acetals, and special reagents have to be used to assist their hydrolysis and reveal the hidden carbonyl group. Sulfur is less basic than oxygen, so the protonated species is lower in concentration at a given pH, and the sulfur 3p lone pairs are less able to form a stable π bond to carbon than are the oxygen 2p lone pairs.

poor reaction...

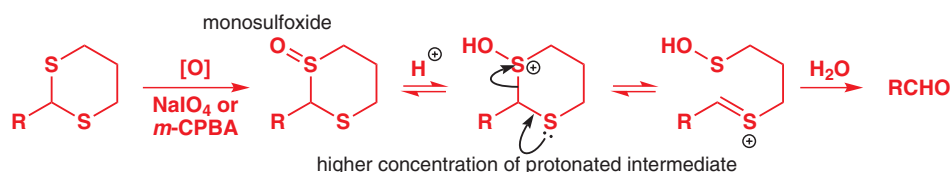


good reaction...

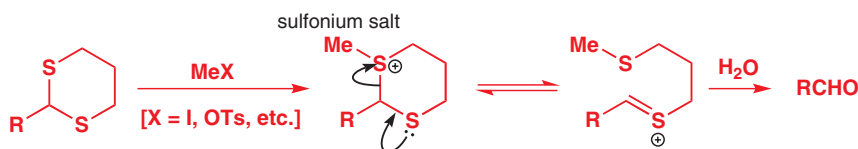


- Sulfur compounds are less basic than oxygen compounds and C=S compounds are less stable than C=O compounds.

The most obvious solution to this problem is to provide a better electrophile than the proton for sulfur. Mercury, Hg(II), is one solution. Another is oxidation of one sulfur to the sulfoxide. Protonation can now occur on the more basic oxygen atom of the sulfoxide and the concentration of the vital intermediate is increased.



A third solution is methylation, since sulfur is a better nucleophile than oxygen for saturated carbon. The sulfonium salt can decompose in the same way to give the free aldehyde. There are many more methods for hydrolysing dithioacetals and their multiplicity should make you suspicious that none is very good.



Hydrogenation of C–S bonds in both sulfides and thioacetals is often achieved with **Raney nickel**, the finely divided form of nickel made by dissolving away the aluminium from a powdered nickel–aluminium alloy using alkali. It can be used either as a catalyst for hydrogenation with gaseous hydrogen or as a reagent since it often contains sufficient adsorbed hydrogen (from the reaction of aluminium with alkali) to effect reductions alone. Thioacetalization followed by Raney nickel reduction is a useful way of replacing a C=O group with CH₂.

➡ Raney nickel was introduced on p. 537.

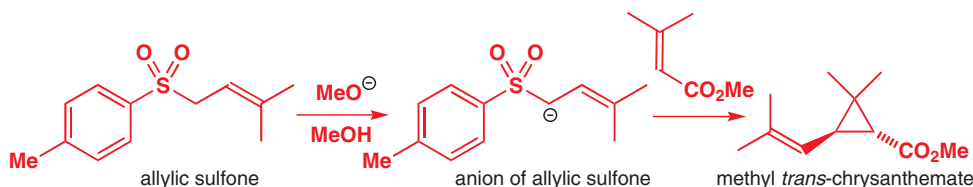
● Dithianes are 'acyl anion equivalents'

A sequence in which a carbonyl group has been masked as a sulfur derivative, alkylated with an electrophile, and then revealed again is a **nucleophilic acylation**. These nucleophilic equivalents of carbonyl compounds are known as **acyl anion equivalents**. In the retrosynthetic terms of Chapter 28 they are d¹ reagents corresponding to the acyl anion synthon.

Anions from sulfones

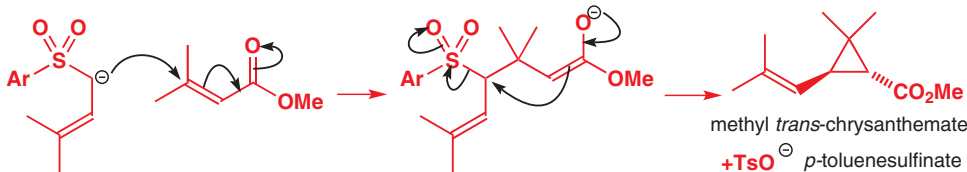
If the sulfur is at a higher oxidation level, it is much easier to make adjacent anions, and sulfones excel in this regard. The allylic sulfone we made earlier in the chapter (p. 659) can be deprotonated and added to an unsaturated ester to give a cyclopropane. Notice how much weaker a base (MeO[−]) is needed here, as the anion is stabilized by sulfone and alkene.

■ You will see more reactions of this sort in which sulfur has a dual role as anion-stabilizing and as leaving group in the section on sulfonium salts.



The first step is conjugate addition of the highly stabilized anion. The intermediate enolate then closes the three-membered ring by favourable nucleophilic attack on the allylic carbon. The leaving group is the sulfinate anion and the stereochemistry comes from the most favourable arrangement in the transition state for this ring closure. The product is the methyl ester of the important chrysanthemic acid found in the natural pyrethrum insecticides.

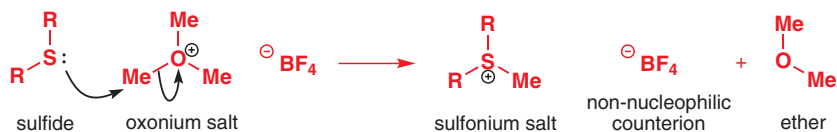
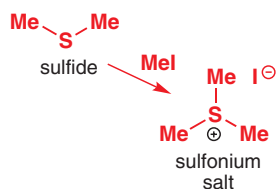
■ In Chapter 22 we established that more stable nucleophiles, and hence more reversible reactions, are likely to favour conjugate addition.



Sulfonium salts

Sulfides are nucleophiles even when not deprotonated—the sulfur atom will attack alkyl halides to form sulfonium salts. This is, of course, a familiar pattern of reactivity for amines, and you have seen phosphonium salts formed in a similar way.

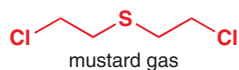
This reaction is an equilibrium and it may be necessary in making sulfonium salts from less reactive sulfides (sterically hindered ones, for example) to use more powerful alkylating agents with non-nucleophilic counterions, for example $\text{Me}_3\text{O}^+ \text{BF}_4^-$, trimethyloxonium fluoroborate (also known as Meerwein's salt). The sulfur atom captures a methyl group from O^+ , but the reverse does not happen and the BF_4^- anion is not a nucleophile. Not only is dimethyl ether a poor nucleophile, it is also a gas and is lost from the reaction mixture. The same principle is used to make sulfides from other sulfides.



The most important chemistry of sulfonium salts is based on one or both of two attributes:

1. Sulfonium salts are electrophiles: nucleophilic substitution displaces a neutral sulfide leaving group.
2. Sulfonium salts can be deprotonated to give sulfonium ylids.

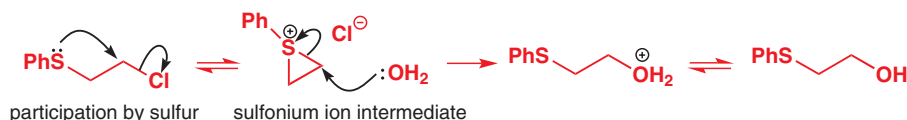
Sulfonium salts as electrophiles



During the First World War, mustard gas was developed as a chemical weapon—it causes the skin to blister and is an intense irritant of the respiratory tract. Its reactivity towards human tissue is related to the following observation and is gruesome testimony to the powerful electrophilic properties of sulfonium ions.



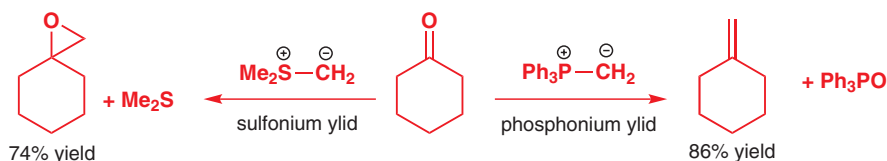
In both cases, intramolecular displacement of the chloride leaving group by the sulfur atom—or, as we should call it, **participation by sulfur**—gives a three-membered cyclic sulfonium ion intermediate (an **episulfonium** or **thiiranium ion**). Nucleophilic attack on this electrophilic sulfonium ion, either by water or by the structural proteins of the skin, is very fast. Of course, mustard gas can react twice in this way. You will see several more examples of reactions in which a sulfonium ion intermediate acts as an electrophile in the next section.



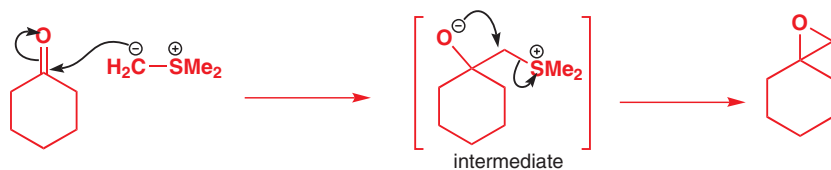
Sulfonium ylids

The positive charge carried by the sulfur atom means that the protons next to the sulfur atom in a sulfonium salt are significantly more acidic than those in a sulfide, and sulfonium salts can be deprotonated to give **sulfonium ylids**.

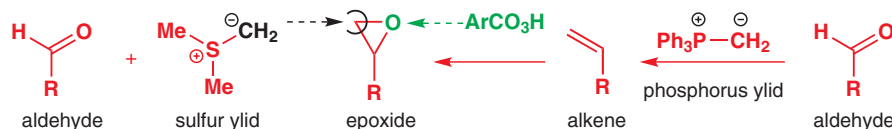
In Chapter 11 we discussed the Wittig reaction of phosphonium ylids with carbonyl compounds. Sulfonium ylids react with carbonyl compounds too, but in quite a different way—compare these two reactions.



Phosphonium ylids give alkenes while sulfonium ylids give epoxides. Why should this be the case? The driving force in the Wittig reaction is formation of the strong P=O bond—that force is much less in the sulfur analogues (the P=O bond strength in Ph_3PO is 529 kJ mol^{-1} ; in Ph_2SO the S=O bond strength is 367 kJ mol^{-1}). The first step is the same in both reactions: the carbanion of the ylid attacks the carbonyl group in a nucleophilic addition reaction. The intermediate in the Wittig reaction cyclizes to give a four-membered ring but this does not happen with the sulfur ylids. Instead, the intermediate decomposes by intramolecular nucleophilic substitution of Me_2S by the oxyanion.

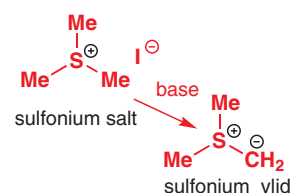


Sulfonium ylids are therefore useful for making epoxides from aldehydes or ketones; other ways you have met of making epoxides (Chapter 19) started with alkenes that might themselves be made with phosphorus ylids.



Some chemists working on a route to some potential β -blocker drugs needed the epoxide below, and since 4-cyclopropylbenzaldehyde was more readily available than 4-cyclopropyl styrene, they decided to use the aldehyde as the starting material and make the epoxide in one step using a sulfonium ylid.

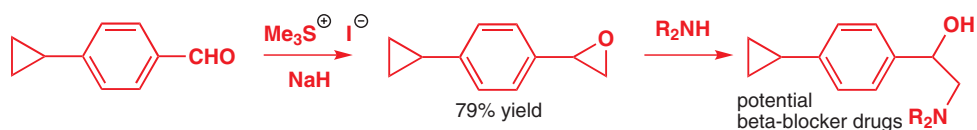
➔ Participation is discussed in detail in Chapter 36.



■ A reminder. An **ylid** is a species with positive and negative charges on adjacent atoms.

➔ The Wittig reaction of phosphonium ylids was introduced on p. 237 and appears again at the end of this chapter, p. 689.

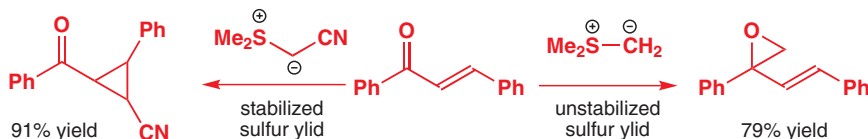
🖱 Interactive mechanism for epoxide formation using sulfonium ylids



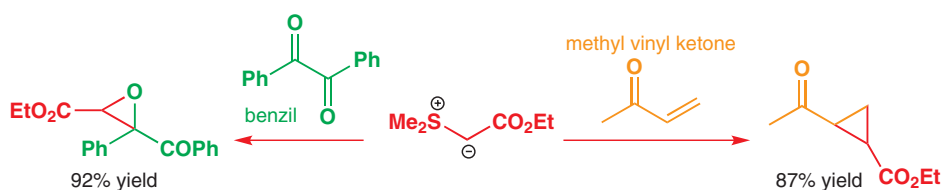
'Stabilized' sulfonium ylids

If there is a conjugating group on the carbanion carbon atom of the ylid, the ylid is more stable and its reactions may change. Firstly, an example where the ylid is stabilized by a cyanide. As you have just seen, the simple sulfonium ylid gives the epoxide from this α,β -unsaturated ketone. But the 'stabilized' ylid gives the cyclopropane instead.

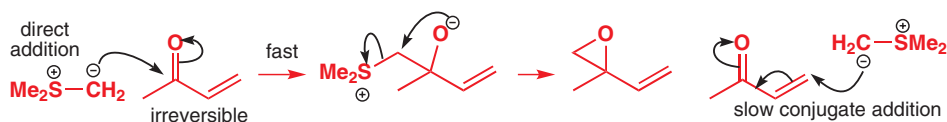
➡ You saw similar reactions of enolates to form epoxides (the Darzens reaction) and cyclopropanes in Chapter 26.



In the absence of the conjugated alkene, both types of ylid give epoxides—the ester-stabilized ylid, for example, reacts with the diketone known as benzil to give an epoxide but with methyl vinyl ketone (but-3-en-2-one) to give a cyclopropane.

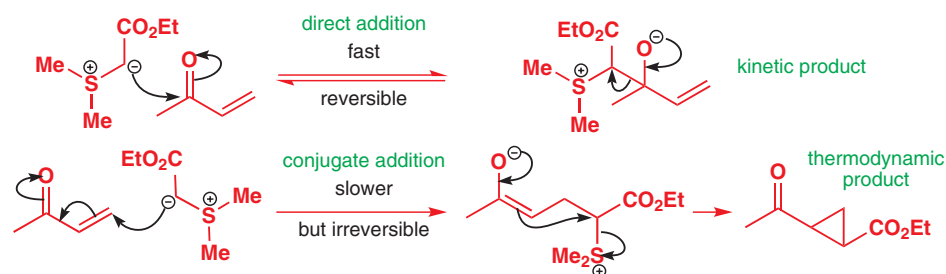


Why does the stabilized ylid prefer to react with the double bond? The enone has two electrophilic sites, but from Chapter 22, in which we discussed the regioselectivity of attack of nucleophiles on Michael acceptors like this, you would expect that direct 1,2-attack on the ketone is the faster reaction. This step is irreversible, and subsequent displacement of the sulfide leaving group by the alkoxide produces an epoxide. Whether a cyclopropane product would have been more stable is irrelevant to the outcome: the epoxide forms faster and is therefore the kinetic product.



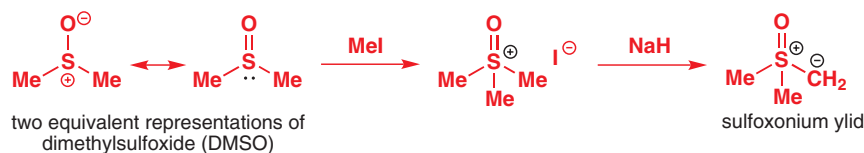
With a stabilized ylid, direct addition to the carbonyl group is, in fact, probably still the faster reaction. But in this case, the starting materials are sufficiently stable that the reaction is reversible, and the sulfonium ylid is re-expelled before the epoxide has a chance to form. Meanwhile, some ylid adds to the ketone in a 1,4 (Michael or conjugate) fashion. 1,4-Addition, although slower, is energetically more favourable because the new C–C bond is gained at the expense of a (relatively) weak C=C π bond rather than a (relatively) strong C=O π bond, and is therefore irreversible. Eventually, all the ylid ends up adding in a 1,4-fashion, generating an enolate as it does so, which cyclizes to give the cyclopropane, which is the thermodynamic product. This is another classic example of kinetic versus thermodynamic control, and you can add it to your mental list of examples.

➡ Other examples can be found on pp. 266 and 505.

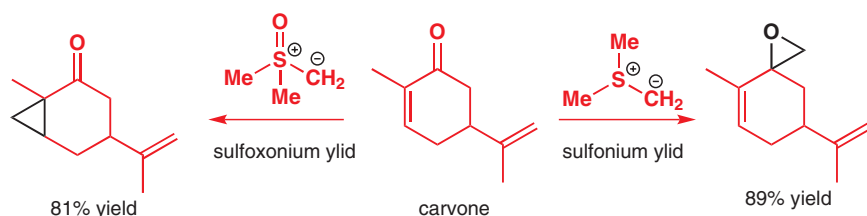


Sulfoxonium ylids

There is another, very important, class of stabilized sulfur ylids that owe their stability not to an additional anion-stabilizing substituent but to a more anion-stabilizing sulfur group. These are the **sulfoxonium ylids**, made from dimethylsulfoxide by S_N2 substitution with an alkyl halide. Note that the sulfur atom is the nucleophile rather than the oxygen atom in spite of the charge distribution. The high-energy sulfur lone pair is better at S_N2 substitution at saturated carbon—a reaction that depends very little on charge attraction (Chapter 15).



Sulfoxonium ylids react with unsaturated carbonyl compounds in the same way as the stabilized ylids that you have met already—they form cyclopropanes rather than epoxides. The example below shows one consequence of this reactivity pattern—by changing from a sulfonium to a sulfoxonium ylid, high yields of either epoxide or cyclopropane can be formed from an unsaturated carbonyl compound (this one is the terpene known as carvone).

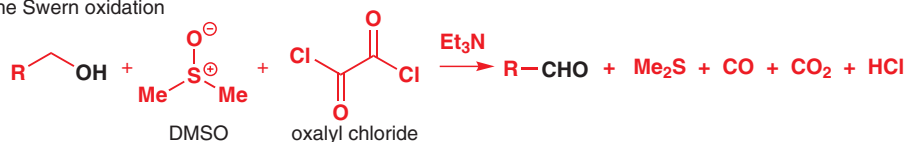


Interactive mechanism for three-membered ring formation with sulfonium and sulfoxonium ylids

The Swern oxidation

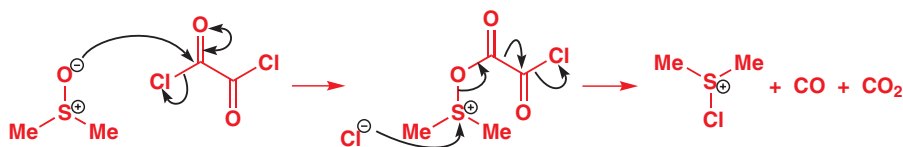
This important reaction featured briefly in Chapter 23 as an important method of oxidizing alcohols to aldehydes. We said there that we would discuss this interesting reaction later and now is the time.

the Swern oxidation



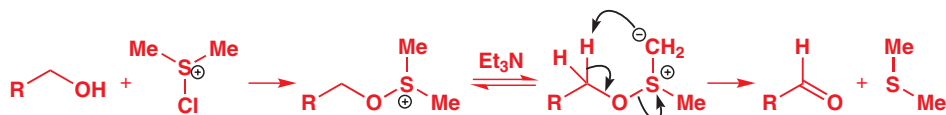
➡ Turn to p. 545 for a comparison of the Swern oxidation with other similar methods.

In the first step, DMSO reacts with oxalyl chloride to give an electrophilic sulfur compound. You should not be surprised that it is the charged oxygen atom that attacks the carbonyl group rather than the soft sulfur atom. Chloride is released in this acylation and it attacks the positively charged sulfur atom, expelling a remarkable leaving group that fragments into three pieces: CO_2 , CO , and a chloride ion. Entropy favours this reaction.



The alcohol has been a spectator in these events so far but the chlorosulfonium ion now formed can react with it to give a new sulfonium salt. This is the sole purpose of all the reactions up to this point. This sulfonium salt is deprotonated by the base (Et₃N) to form an ylid. The final step completes the redox reaction: the transfer of a proton to the anionic carbon gives an aldehyde, with overall reduction of dimethylsulfoxide (DMSO) to dimethylsulfide (DMS).

Interactive mechanism of the Swern oxidation of alcohols



➡ In Chapter 35 you will learn to call this last step a pericyclic reaction.

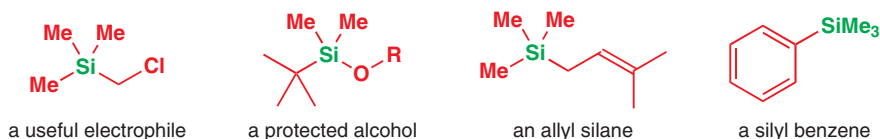
Silicon and carbon compared

Silicon is immediately below carbon in the periodic table and the most obvious similarity is that both elements normally have a valency of four and both form tetrahedral compounds. There are important differences in the chemistry of carbon and silicon—silicon is less important and many books are devoted solely to carbon chemistry but relatively few to silicon chemistry. Carbon forms many stable trigonal and linear compounds containing π bonds; silicon forms few. The most important difference is the strength of the silicon–oxygen σ bond (368 kJ mol⁻¹) and the relative weakness of the silicon–silicon (230 kJ mol⁻¹) bond. Together these values account for the absence, in the oxygen-rich atmosphere of earth, of silicon analogues of the plethora of structures possible with a carbon skeleton.

Average bond energies, kJ mol⁻¹

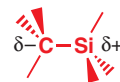
X	H–X	C–X	O–X	F–X	Cl–X	Br–X	I–X	Si–X
C	416	356	336	485	327	285	213	290
Si	323	290	368	582	391	310	234	230
ratio	1.29	1.23	0.91	0.83	0.84	0.92	0.91	1.26

Several of the values in the table give insight into the reactivity differences between carbon and silicon. Bonds to electronegative elements are generally stronger with silicon than with carbon; in particular, the silicon–fluorine bond is one of the strongest single bonds known, while bonds to electropositive elements are weaker. Silicon–hydrogen bonds are much weaker than their carbon counterparts and can be cleaved easily. Here are a few organosilicon compounds.



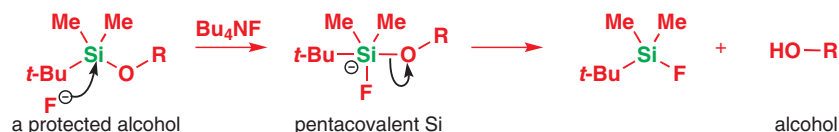
In this section we will mostly discuss compounds with four Si–C bonds. Three of these bonds will usually be the same so we will often have a Me₃Si group attached to an organic molecule. We shall discuss reactions in which something interesting happens to the organic molecule as one of the Si–C bonds reacts to give a new Si–F or Si–O bond. We shall also discuss organosilicon compounds as reagents, such as triethylsilane (Et₃SiH), which is a reducing agent whereas Et₃C–H is not.

The carbon–silicon bond is strong enough for the trialkyl silyl group to survive synthetic transformations on the rest of the molecule but weak enough for it to be cleaved specifically when we want. In particular, fluoride ion is a poor nucleophile for carbon compounds but attacks silicon very readily. Another important factor is the length of the C–Si bond (1.89 Å)—it is significantly longer than a typical C–C bond (1.54 Å). Silicon has a lower electronegativity (1.8) than carbon (2.5) and therefore C–Si bonds are polarized towards the carbon. This makes the silicon susceptible to attack by nucleophiles. The strength of the C–Si bond means that alkyl silanes are stable but the most useful chemistry arises from carbon substituents other than simple alkyl groups.



Silicon has an affinity for electronegative atoms

The most effective nucleophiles for silicon are the electronegative ones that will form strong bonds to silicon. Those based on oxygen or halide ions (chloride and fluoride) are pre-eminent. You saw this in the choice of reagent for the selective cleavage of silyl ethers in Chapter 23. Tetrabutylammonium fluoride is often used as this is an organic-soluble ionic fluoride and forms a silyl fluoride as the by-product. The mechanism is not a simple S_N2 process and has no direct analogue in carbon chemistry. It looks like a substitution at a hindered tertiary centre, which ought to be virtually impossible. Two characteristics of silicon facilitate the process: the long silicon–carbon bonds relieve the steric interactions and the d orbitals of silicon provide a target for the nucleophile that does not have the same geometric constraints as a C–O σ^* orbital.

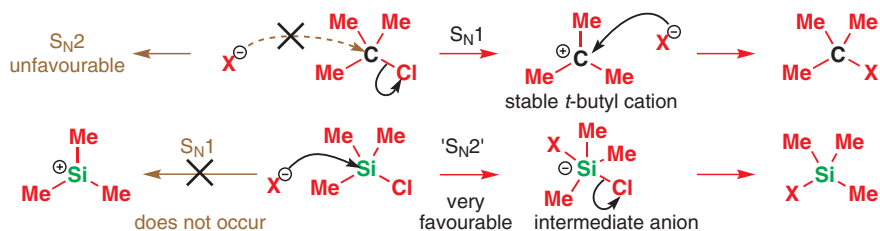


Attack of the fluoride on the empty d orbital leads to a negatively charged pentacoordinate intermediate that breaks down with loss of the alkoxide. The discrete pentacoordinate trigonal bipyramid intermediate contrasts with the similarly shaped pentacoordinate transition state of a carbon-based S_N2 reaction. It is often omitted in mechanistic schemes because it is formed slowly and decomposes quickly, and the mechanism is still referred to as ' S_N2 at silicon'.

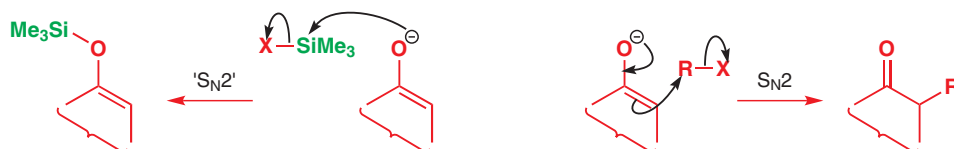
- Silicon forms strong bonds with oxygen and very strong bonds with fluorine.

Nucleophilic substitution at silicon

You may wonder why trimethylsilyl chloride does not use the S_N1 mechanism familiar from the analogous carbon compound *t*-butyl chloride. There is, in fact, nothing wrong with the Me_3Si^+ cation—it can be detected in mass spectra, for example. The reason is simply that the ' S_N2 ' reaction at silicon is too good for S_N1 to compete.



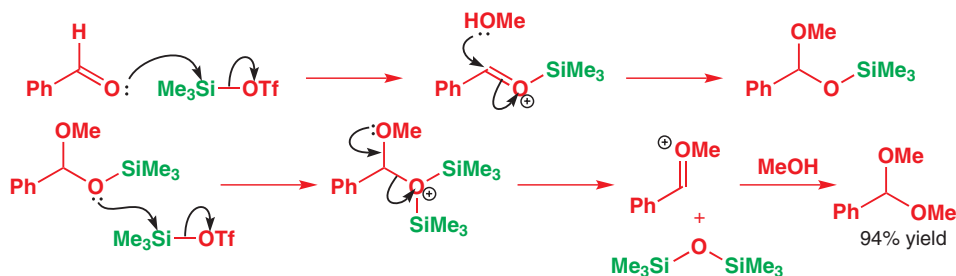
There are some important differences between the S_N2 substitutions at Si and at C. Alkyl halides are soft electrophiles but silyl halides are hard electrophiles. Alkyl halides react only very slowly with fluoride ion but silyl halides react more rapidly with fluoride than with any other nucleophile. The best nucleophiles for saturated carbon are neutral or based on elements down the periodic table (S, Se, I) or both. The best nucleophiles for silicon are charged and based on highly electronegative atoms (chiefly F, Cl, and O). A familiar example is the reaction of enolates at carbon with alkyl halides but at oxygen with silyl chlorides (Chapter 20).



When a Me_3Si group is removed from an organic molecule with hydroxide ion, the product is not the silanol as you might expect but the silyl ether 'hexamethyldisiloxane'.



The other side of the coin is that the $\text{S}_{\text{N}}2$ reaction at carbon is *not* much affected by partial positive charge (δ^+) on the carbon atom. The ' $\text{S}_{\text{N}}2$ ' reaction at silicon *is* affected by the charge on silicon. The most electrophilic silicon compounds are the silyl triflates and it is estimated that they react some 10^8 – 10^9 times faster with oxygen nucleophiles than do silyl chlorides. Trimethylsilyl triflate is, in fact, an excellent Lewis acid and can be used to form acetals or silyl enol ethers from carbonyl compounds, and to react these two together in aldol-style reactions. In all three reactions the triflate attacks an oxygen atom. In the acetal formation, silylation occurs twice at the carbonyl oxygen atom and the final leaving group is hexamethyldisiloxane. You should compare this with the normal acid-catalysed mechanism described in Chapter 11, where the carbonyl group is twice protonated and the leaving group is water.



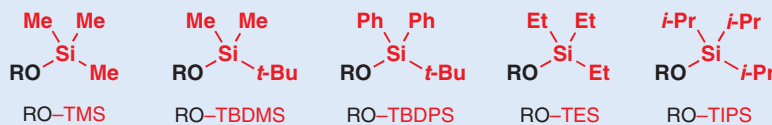
Silyl ethers are versatile protecting groups for alcohols

Silicon-based protecting groups for alcohols are the best because they are the most versatile. They are removed by nucleophilic displacement with fluoride or oxygen nucleophiles and the rate of removal depends mostly on the steric bulk of the silyl group. The simplest is trimethylsilyl (Me_3Si or often just TMS), which is also the most easily removed as it is the least hindered. In fact, it is removed so easily by water with a trace of base or acid that special handling is required to keep this labile group in place. These protecting groups are discussed in Chapter 23.

Replacement of the one of the methyl groups with a much more sterically demanding tertiary butyl group gives the *t*-butyldimethylsilyl (TBDMS) group, which is stable to normal handling and survives aqueous work-up or column chromatography on silica gel. The stability to these isolation and purification conditions has made TBDMS (sometimes over-abbreviated to TBS) a very popular choice for organic synthesis. TBDMS is introduced by a substitution reaction on the corresponding silyl chloride with imidazole in DMF. Yields are usually virtually quantitative and the conditions are mild. Primary alcohols are protected in the presence of secondary alcohols. Removal relies on the strong affinity of fluoride for silicon and is usually very efficient and selective.

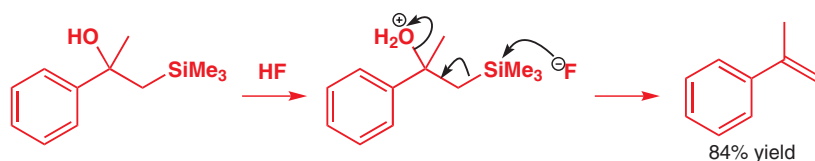
However, a protecting group is useful only if it can be introduced and removed in high yield without affecting the rest of the molecule and if it can survive a wide range of conditions in the course of the synthesis. The extreme steric bulk of the *t*-butyldiphenylsilyl (TBDPS) group makes it useful for selective protection of unhindered primary alcohols in the presence of secondary alcohols.

The most stable common silyl protecting group (triisopropylsilyl or TIPS) has three branched alkyl substituents to protect the central silicon from attack by nucleophiles, which would lead to cleavage. All three hindered silyl groups (TBDMS, TBDPS, and TIPS) have excellent stability but can still be removed with fluoride.



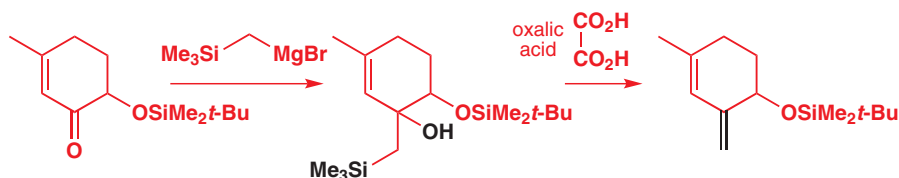
The Peterson elimination

There are many reactions in organic chemistry in which an Me_3Si group acts like a proton. Just as acidic protons are removed by bases, silicon is readily removed by hard nucleophiles, particularly F^- or RO^- , and this can promote an elimination. An example is shown here.



Interactive mechanism for Peterson elimination

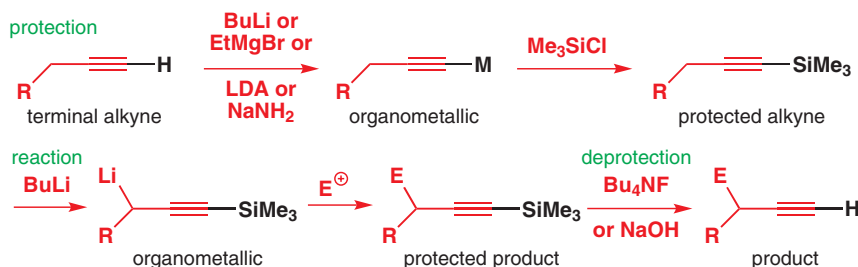
This reaction is known as the **Peterson elimination**. It is rather like those we discussed in Chapter 17—eliminations of alcohols under acidic conditions to give alkenes. But, unlike those reactions, it is fully regioselective and so is particularly useful for making double bonds where other elimination methods might give the wrong regioisomer or mixtures of regioisomers. In this next example only one product is formed, in high yield, and it has an exocyclic double bond. Just think what would have happened without the silicon atom (ignore the one attached to the oxygen—that's just a protecting group). This compound is, in fact, an intermediate in a synthetic route to the important anticancer compound Taxol.



The Peterson reaction is particularly useful for making terminal or exocyclic double bonds connectively because the starting material (the magnesium derivative shown above) is easily made from available $\text{Me}_3\text{SiCH}_2\text{Br}$.

Alkynyl silanes are used for protection and activation

Terminal alkynes have an acidic proton ($\text{p}K_a$ ca. 25) that can be removed by very strong bases such as organometallic reagents (Grignards, RLi , etc.). While this is often what is intended, in other circumstances it may be an unwanted side reaction that would consume an organometallic reagent or interfere with the chosen reaction. Exchange of the terminal proton of an alkyne for a trimethylsilyl group exploits the relative acidity of the proton and provides a neat solution to these problems. The SiMe_3 group protects the terminus of the alkyne during the reaction but can then be removed with fluoride or sodium hydroxide. A classic case is the removal of a proton next door to a terminal alkyne.

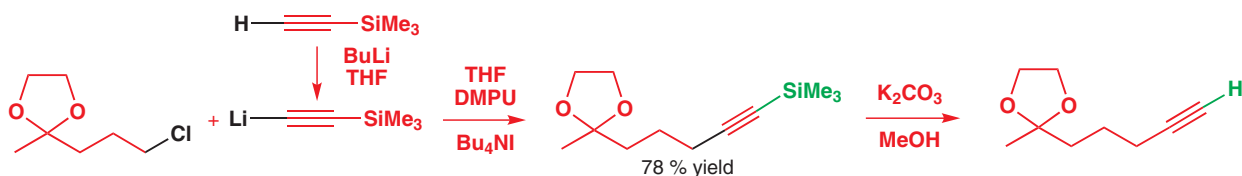


■ The position 'next door' to an alkyne is sometimes called a 'propargylic' position. Propargyl alcohol is $\text{HC}\equiv\text{CCH}_2\text{OH}$.

➡ Alkynyl lithiums and Grignards were made in this way in Chapter 9.

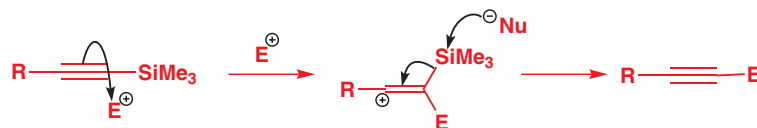
Additionally, acetylene itself is a useful two-carbon building block but is not very convenient to handle as it is an explosive gas. Trimethylsilylacetylene is a distillable liquid that is a convenient substitute for acetylene in reactions involving the lithium derivative as it has only one acidic proton. The synthesis of this alkynyl ketone is an example. Deprotonation with butyl lithium provides the alkynyl lithium that reacted with the alkyl chloride in the presence of iodide as nucleophilic catalyst (see Chapter 15). Removal of the trimethylsilyl

group with potassium carbonate in methanol allowed further reaction on the other end of the alkyne.



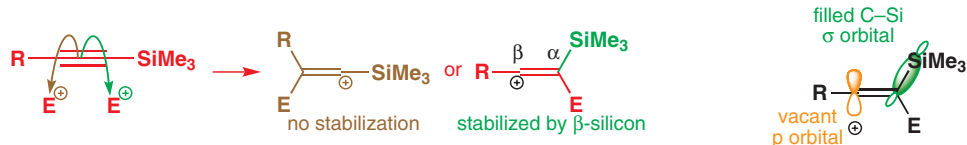
Silicon stabilizes a positive charge on the β carbon

In common with ordinary alkynes, silylated alkynes are nucleophilic towards electrophiles. The presence of the silicon has a dramatic effect on the regioselectivity of this reaction: attack occurs only at the atom directly bonded to the silicon. This must be because the intermediate cation is stabilized.



➡ This was explained in Chapter 15.

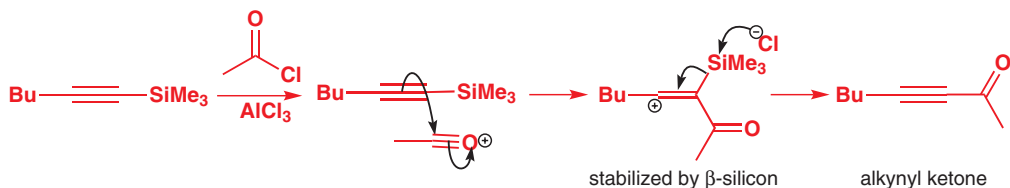
The familiar hierarchy of carbocation stability—tertiary > secondary > primary—is due to the stabilization of the positive charge by donation of electron density from adjacent C–H or C–C bonds (their filled σ orbitals to be precise) that are aligned correctly with the vacant orbital. The electropositive nature of silicon makes C–Si bonds even more effective donors: a silyl group β to a positive charge (i.e. attached to the next-door carbon) stabilizes a positive charge so effectively that the course of a reaction involving cationic intermediates is often completely controlled. This is stabilization by σ donation.



■ The nucleophile does not need to be very powerful because of the weakening of the C–Si bond. Many neutral molecules with a lone pair and almost any anion will do, even triflate ($\text{CF}_3\text{SO}_2\text{O}^-$).

The stabilization of the cation also weakens the C–Si bond by delocalization so that the bond is more easily broken. Attack of a nucleophile (particularly a halogen or oxygen nucleophile) on silicon removes it from the organic fragment and the net result is electrophilic substitution in which the silicon has been replaced by the electrophile.

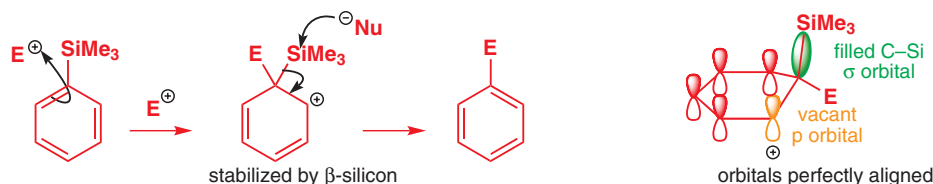
This is useful for the synthesis of alkynyl ketones, which are difficult to make directly with conventional organometallic reagents such as alkynyl–Li or –MgBr because they add a second time to the ketone product. Alkynyl silanes react in a Friedel–Crafts manner with acid chlorides in the presence of Lewis acids, such as aluminium chloride, to give the ketones.



Aryl silanes undergo *ipso* substitution with electrophiles

Exactly the same sort of mechanism accounts for the reactions of aryl silanes with electrophiles under Friedel–Crafts conditions. Instead of the usual rules governing *ortho*, *meta*, and *para* substitution using the directing effects of the substituents, there is just one rule: the silyl group is replaced by the electrophile at the same atom on the ring—this is known as *ipso*

substitution. Actually, this selectivity comes from the same principles as those used for ordinary aromatic substitution (Chapter 21): the electrophile reacts to produce the most stable cation—in this case β to silicon. Cleavage of the weakened C–Si bond by any nucleophile leads directly to the *ipso* product.

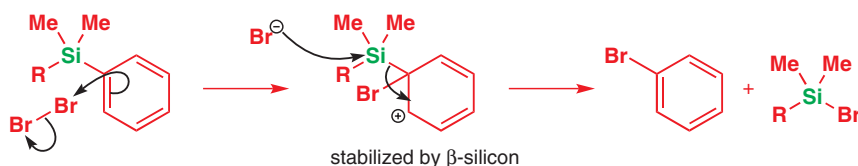


There is an alternative site of attack *meta* to silicon that would lead to a cation β to Si. But this cation is not particularly stable because the vacant p orbital is orthogonal to the C–Si bond and so cannot interact with it as the C–Si bond is still in the plane of the ring. This illustrates that it is more important to understand the origin of the effect based on molecular orbitals rather than simply to remember the result.

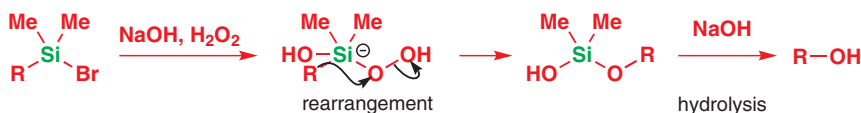


This reactivity of aryl silanes is used to convert stable phenyldimethylsilyl compounds into more reactive compounds such as alcohols by a reaction such as that shown in the margin. Several reagents can be used, all of which induce *ipso* substitution of the phenyl silane. The reaction with bromine is typical. Bromobenzene is produced together with a silyl bromide that is activated towards subsequent oxidation.

The mechanism of electrophilic desilylation is the same as that for electrophilic aromatic substitution except that the proton is replaced by the trimethylsilyl group. The silicon stabilizes the intermediate cation, and hence the transition state leading to it, to such an extent that the rate is many orders of magnitude faster. This is the first step with bromine.



The rest of the reaction sequence involves displacement of Br^- by HOO^- , addition of hydroxide, rearrangement, and hydrolysis.

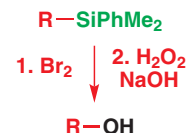
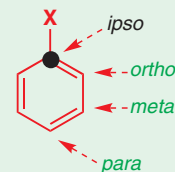


● Trimethylsilyl and other silyl groups stabilize a positive charge on a β carbon and are lost very easily. They can be thought of as very reactive protons or ‘super protons’.

Vinyl silanes offer a regio- and stereoselective route to alkenes

Vinyl silanes react with electrophiles in a similarly regioselective process in which the silicon is replaced by the electrophile at the *ipso* carbon atom. The stereochemistry of the vinyl silane is important because this exchange usually occurs with retention of geometry as well.

■ The Latin word *ipso* means ‘itself’—the *self same* site as that occupied by the SiR_3 group. A reminder:

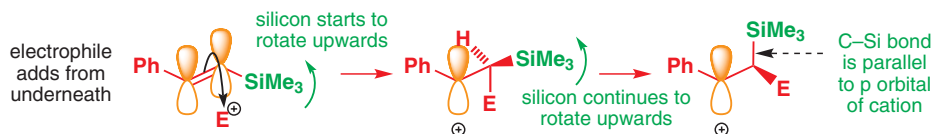


■ A group of similar Si to OH conversions are known as Fleming–Tamao oxidations, after the two independent discoverers.

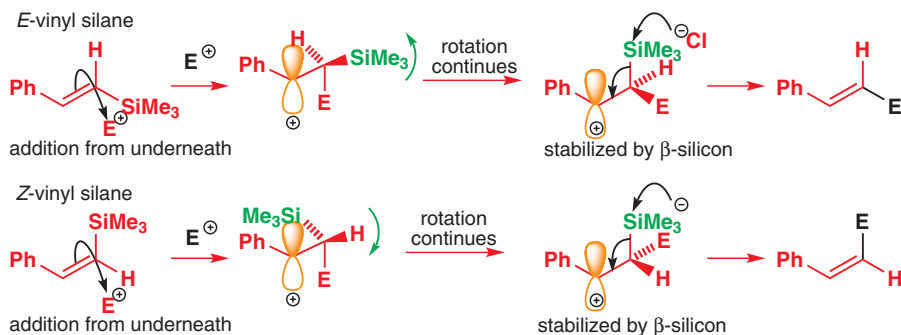
➡ This mechanism should remind you of the mechanism of the oxidation of boranes, which often follows on from hydroboration—see Chapter 19, p. 446.



This is a curious and interesting reaction that deserves explanation. Addition of the electrophile next to silicon leads to the more stable cation β to silicon. In the vinyl silane the C–Si bond is orthogonal to the p orbitals of the π bond, but as the electrophile attacks the π bond, say from underneath, the Me_3Si group starts to move upwards. As it rotates, the angle between the C–Si bond and the remaining p orbital decreases from 90° . As the angle decreases, the interaction between the C–Si bond and the empty p orbital of the cation increases. There is every reason for the rotation to continue in the same direction and no reason for it to reverse. The diagram shows that, in the resulting cation, the electrophile is in the position formerly occupied by the Me_3Si group, *trans* to Ph. Loss of the group now gives retention of stereochemistry.



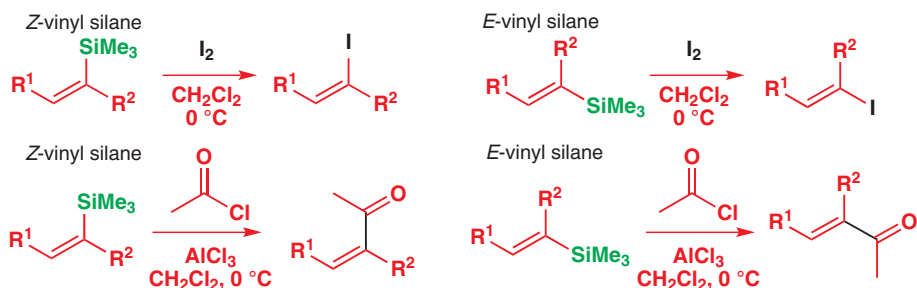
The intermediate cation has only a single bond and so rotation might be expected to lead to a mixture of geometrical isomers of the product but this is not observed. The bonding interaction between the C–Si bond and the empty p orbital means that rotation is restricted. This stabilization weakens the C–Si bond and the silyl group is quickly removed before any further rotation can occur. The stabilization is effective only if the C–Si bond is correctly aligned with the vacant orbital, which means it must be in the same plane—rather like a π bond. Here is the result for both *E* and *Z* isomers of the vinyl silane.



Interactive mechanism for reaction of vinyl silanes with electrophiles

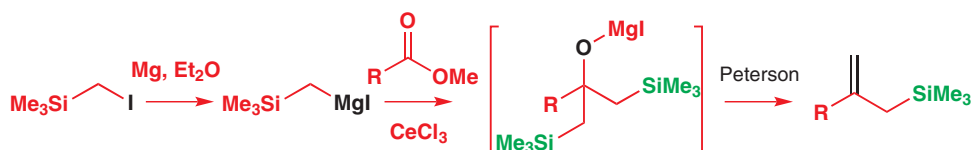
Geometrically pure vinyl halides are important starting materials for transition-metal-catalysed alkene synthesis (Chapter 40).

It is unusual for silicon to be required in the final product of a synthetic sequence and the stereospecific removal of silicon from vinyl silanes makes them useful reagents that can be regarded as rather stable vinylic organometallic reagents that will react with powerful electrophiles, preserving the double bond location and geometry. Protodesilylation, as the process of replacing silicon with a proton is known, is one such important reaction. The halogens are also useful electrophiles while organic halides, particularly acid chlorides, in the presence of Lewis acids, form vinyl halides and unsaturated ketones of defined geometry.

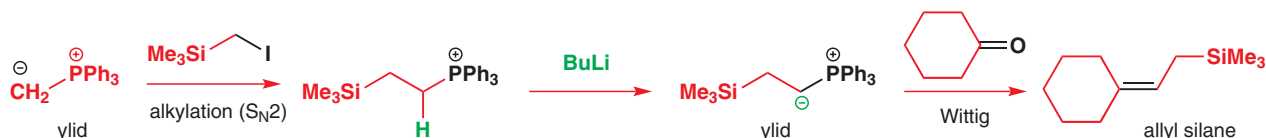


Allyl silanes as nucleophiles

If the silyl group is moved along the carbon chain by just one atom, an allyl silane results. Allyl silanes can be produced from allyl organometallic reagents but there is often a problem over which regioisomer is produced and mixtures often result. Better methods control the position of the double bond. Two useful examples take advantage of the Wittig reaction and the Peterson elimination to construct the alkene linkage. The reagents are prepared from trimethylsilyl halides either by formation of the corresponding Grignard reagent or alkylation with a primary Wittig reagent and deprotonation to form a new ylid. The Grignard reagent, with added cerium trichloride, adds twice to esters to give the corresponding tertiary alcohol, which loses one of its Me_3Si groups in a Peterson elimination to reveal the remaining Me_3Si group as part of allyl silane.



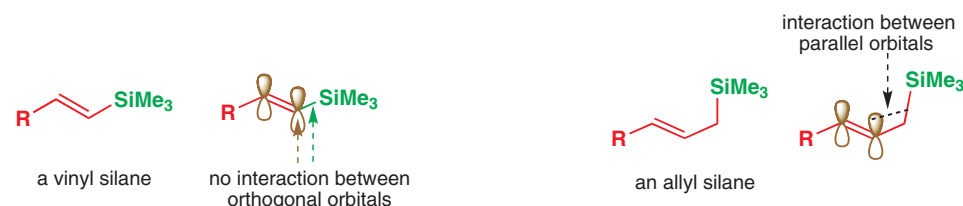
The Wittig reagent is made by alkylation of the simplest ylid with the same silicon reagent. Notice that the leaving group (iodide) is on the carbon next to silicon, not on the silicon itself. Anion formation occurs next to phosphorus because Ph_3P^+ is much more anion-stabilizing than Me_3Si . The ylid reacts with carbonyl compounds such as cyclohexanone in the usual way to produce the allyl silane with no ambiguity over which end of the allyl system is silylated.



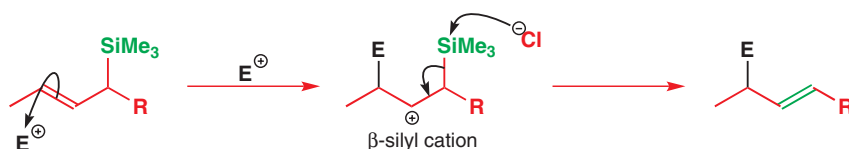
The carbon–silicon bond has two important effects on the adjacent alkene. The presence of a high-energy filled σ orbital of the correct symmetry to interact with the π system produces an alkene that is more reactive with electrophiles, due to the higher-energy HOMO, and the same σ orbital stabilizes the carbocation if attack occurs at the remote end of the alkene. This lowers the transition state for electrophilic addition and makes allyl silanes much more reactive than isolated alkenes.

Allyl silanes are more reactive than vinyl silanes but also react through β -silyl cations

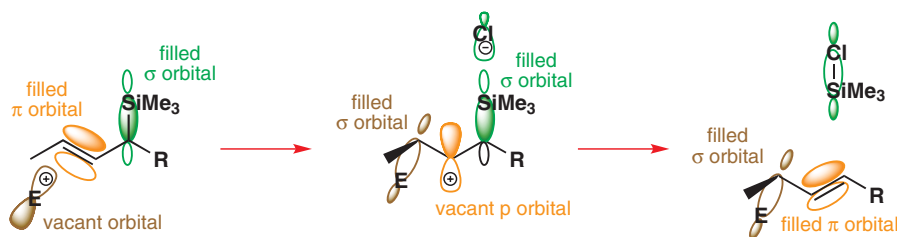
Vinyl silanes have C–Si bonds orthogonal to the p orbitals of the alkene—the C–Si bond is in the nodal plane of the π bond—so there can be no interaction between the C–Si bond and the π bond. Allyl silanes, by contrast, have C–Si bonds that can be, and normally are, parallel to the p orbitals of the π bond so that interaction is possible.



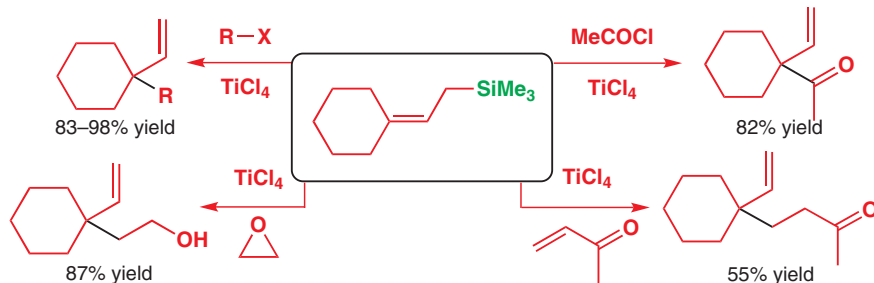
Allyl silanes react with electrophiles with even greater regioselectivity than that of vinyl silanes. The cation β to the silyl group is again formed but there are two important differences. Most obviously, the electrophile attacks at the other end of the allylic system and there is no rotation necessary as the C–Si bond is already in a position to overlap efficiently with the intermediate cation. The process is terminated by loss of silicon in the usual way to regenerate an alkene.



Molecular orbitals demonstrate the smooth transition from the allyl silane, which has a π bond and a C–Si σ bond, to the allylic product with a new π bond and a new σ bond to the electrophile. The intermediate cation is mainly stabilized by σ donation from the C–Si bond into the vacant p orbital but it has other σ -donating groups (C–H, C–C, and C–E) that also help. The overall process is electrophilic substitution with allylic rearrangement. Both the site of attachment of the electrophile and the position of the new double bond are dictated by the silicon.



Allyl silanes are rather like silyl enol ethers: they react with electrophiles, provided they are activated, for example by a Lewis acid. Titanium tetrachloride is widely used but other successful Lewis acids include boron trifluoride, aluminium chloride, and trimethylsilyl triflate. Electrophiles include acylium ions produced from acid chlorides, carbocations from tertiary halides or secondary benzylic halides, activated enones, and epoxides all in the presence of Lewis acid. In each case the new bond is highlighted in black.



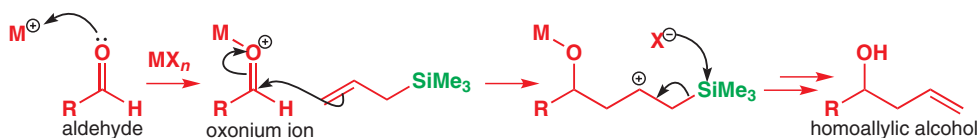
• β -Silyl cations are important intermediates

Vinyl and aryl silanes react with electrophiles at the same (*ipso* or α) atom occupied by silicon. Allyl silanes react at the end of the alkene furthest from silicon (γ). In both cases a β -silyl cation is an intermediate.

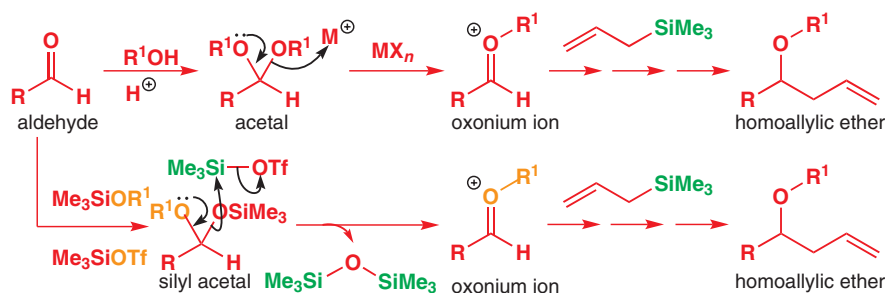
Lewis acids promote couplings via oxonium ions

■ Homoallylic means allylic plus one carbon.

Allyl silanes will also attack carbonyl compounds when they are activated by coordination of the carbonyl oxygen atom to a Lewis acid. The Lewis acid, usually a metal halide such as TiCl_4 or ZnCl_2 , activates the carbonyl compound by forming an oxonium ion with a metal–oxygen bond. The allyl silane attacks in the usual way and the β -silyl cation is desilylated with the halide ion. Hydrolysis of the metal alkoxide gives a homoallylic alcohol.

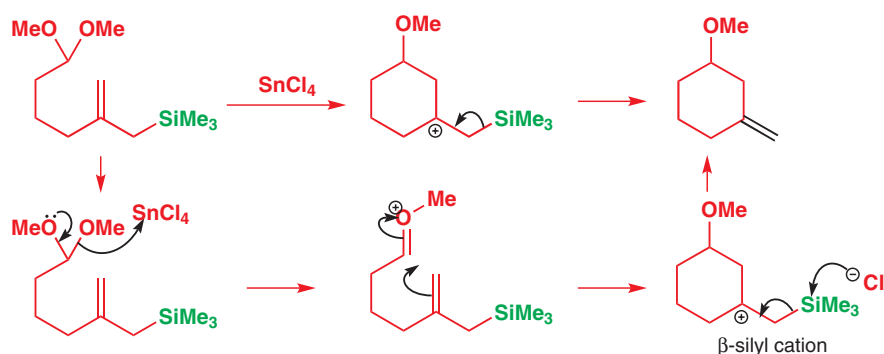


A closely related reactive oxonium ion can be prepared by Lewis acid catalysed breakdown of the corresponding acetal. Alternatively, especially if the acetal is at least partly a silyl acetal, the same oxonium ion can be produced *in situ* using yet more silicon in the form of TMSOTf as the Lewis acid catalyst. All these intermediate oxonium ions act as powerful electrophiles towards allyl silanes, producing homoallylic alcohols or ethers.



■ Note how the Me₃Si group mimics the behaviour of a proton even to the extent of producing (Me₃Si)₂O—the silicon analogue of water.

The regiocontrol that results from using an allyl silane to direct the final elimination is illustrated by this example of an intramolecular reaction on to an acetal promoted by tin tetrachloride. The same reaction can be run in the absence of silicon but the intermediate cation can then lose a range of protons to produce five different products!



The selective synthesis of alkenes

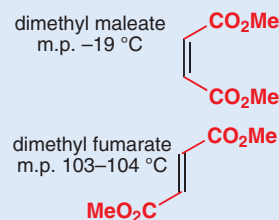
S, Si, P, and other main group elements have several important functions in organic chemistry, and one in which all of S, Si, and P each play a star role is in the synthesis of alkenes. You have met alkenes participating in reactions in a number of chapters, but our discussion of how to *make* alkenes has so far been quite limited. Chapter 17 was about elimination reactions, and there you met E1 and E2 reactions. You had a glimpse of the importance of phosphorus in alkene synthesis in Chapter 11, where you met the Wittig reaction, and earlier in this chapter you saw silicon participating in the Peterson elimination. We're now going to look at related reactions in more detail, addressing especially how to form alkenes with control over their *geometry*. First we need to establish that this is an important task and remind you of some reactions you have already met that can be used for it.

The properties of alkenes depend on their geometry

Geometrical isomers of alkenes are different compounds with different physical, chemical, and biological properties. They are often hard to separate by chromatography or distillation, so it is important that chemists have methods for making them as single isomers.

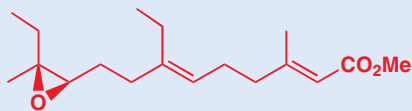
Different physical properties: maleate and fumarate

These two compounds, Z- and E-dimethyl but-2-enedioate, are commonly known as dimethyl maleate and dimethyl fumarate. They provide a telling example of how different the physical properties of geometrical isomers can be. Dimethyl maleate is a liquid with a boiling point of 202°C (it melts at −19°C), while dimethyl fumarate is a crystalline compound with a melting point of 103–104°C.



Different biological properties: juvenile hormone as a pest control

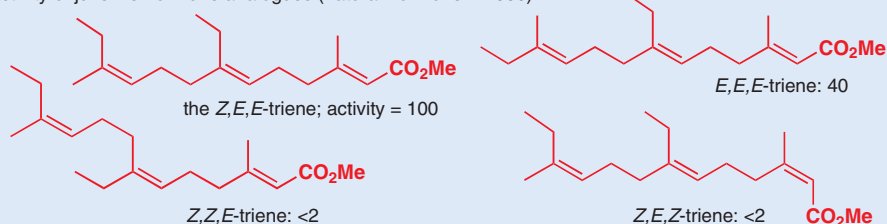
If insect pests can be prevented from maturing they fail to reproduce and can thus be brought under control. Juvenile insects control their development by means of a 'juvenile hormone', one of which is the monoepoxide of a triene:



cecropia juvenile hormone: activity = 1000

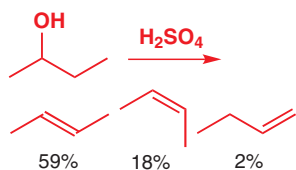
Synthetic analogues of this compound, such as the trienes below, are also effective at arresting insect development, providing that the double bond geometry is controlled. The *Z,E,E* geometrical isomer of the triene is over twice as active as the *E,E,E* isomer, and over 50 times as active as the *Z,Z,E* or *Z,E,Z* isomers.

activity of juvenile hormone analogues (natural hormone = 1000)

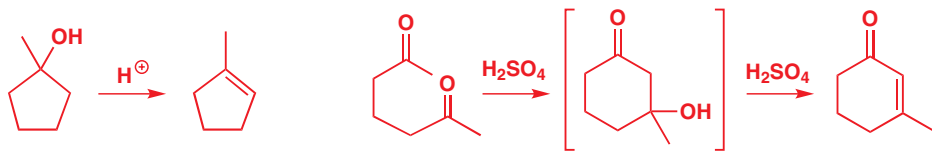


Elimination reactions and stereoselectivity

Unfortunately, most elimination reactions (Chapter 17) offer little control over the geometry of the product: treating *sec*-butanol in acid, for example, gives mainly the more substituted 2-butene, but as a 3:1 mixture of geometrical isomers. But there are some important exceptions. If the product has the double bond inside a ring of less than eight members, it has to be a *cis* double bond. Examples include the simple dehydration of a cyclopentanol and an intramolecular aldol reaction. The six-membered ring is formed before the dehydration step.



■ You should recognize the second of these reactions as the last step of the Robinson annelation you met in Chapter 26.



But how can we use elimination reactions to give single geometrical isomers of open-chain compounds? These reactions fall into four main classes, and we shall look at each in turn before summarizing the most important methods at the end of the chapter.

● Ways of making single geometrical isomers of double bonds

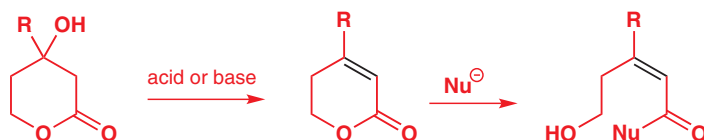
1. Using the fact that only one geometrical isomer is possible (for example, a *cis* double bond in a six-membered ring).
2. The geometrical isomers are in equilibrium and the more stable (usually *E*) is formed.
3. The reaction is stereoselective and the *E* alkene or the *Z* alkene is formed as the main product by kinetic control.
4. The reaction is stereospecific and the alkene geometry depends on the stereochemistry of the starting materials and the mechanism of the reaction.

Exploiting cyclic compounds

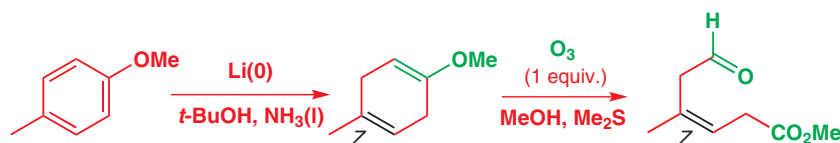
You may think that this method is rather too trivial to be called a method for controlling the geometry of double bonds, as it's only of any use for making cyclic alkenes. Well, chemists are more ingenious than that! It is not necessary to have an all-carbon ring to preserve the *cis* geometry of a double bond. Lactones (cyclic esters) and cyclic anhydrides are useful too.

■ Some people call geometrical isomers diastereoisomers, which they are in a sense: they are stereoisomers that are not mirror images. However, we shall avoid this usage since for most chemists the word *diastereoisomer* carries implications of three-dimensional stereochemistry.

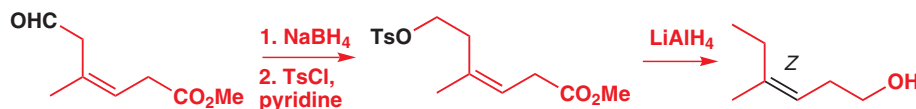
A double bond in a five- or six-membered compound must have a *cis* configuration and compounds like these are readily made. Dehydration of this hydroxylactone can give only a *cis* double bond and ring-opening with a nucleophile (alcohol, hydroxide, amine) gives an open-chain compound also with a *cis* double bond.



E J Corey used a similar idea to make the insect hormone we introduced on p. 678). He realized that the essential *Z* double bond would be easy to make if he were to start with a cyclic molecule (in which only *cis* double bonds are possible) that could be ring-opened to the compound he needed. This is how he did it.

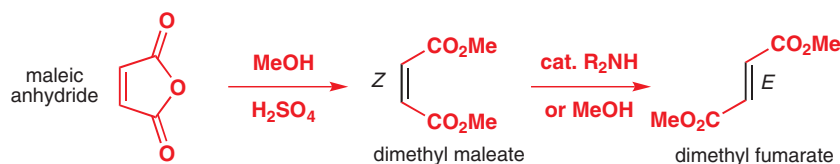


Birch reduction (Chapter 23, p. 542) of a simple aromatic ether generated two *cis* double bonds. The more reactive (because it is more electron-rich) of these reacts with ozone to give an aldehyde-ester in which the *Z* geometry is preserved. NaBH_4 reduces the aldehyde group to a hydroxyl group, which needs to be got rid of: a good way to do this is to tosylate and reduce with LiAlH_4 , which substitutes H for OTs. The LiAlH_4 also does the job of reducing the ester to an alcohol, giving the *Z*-configured compound that Corey needed.



Equilibration of alkenes

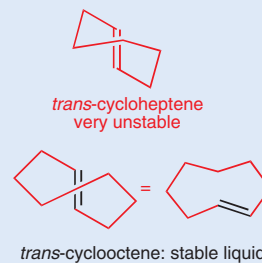
Acyclic *E* alkenes are usually more stable than acyclic *Z* alkenes because they are less sterically hindered. Yet *Z* alkenes do not spontaneously convert to *E* alkenes because the π bond prevents free rotation: the energy required to break the π bond is about 260 kJ mol^{-1} (rotation about a σ bond requires about 10 kJ mol^{-1}). You may therefore find the following result surprising. Dimethyl maleate is easily made by refluxing maleic anhydride in methanol with an acid catalyst. If the product is isolated straight away, a liquid boiling at $199\text{--}202^\circ\text{C}$ is obtained. This is dimethyl maleate. However, if the product is left to stand, crystals of *dimethyl fumarate* (the *E* isomer of dimethyl maleate) form. How has the geometry been inverted so easily?



A clue is that the process is accelerated enormously by a trace of amine. Michael addition of this amine, or of methanol, or any other nucleophile, provides a chemical mechanism by which the π bond can be broken. There is free rotation in the intermediate, and re-elimination of the nucleophile can give either *E* or *Z* alkene. The greater stability and crystallinity of the *E* alkene means that it dominates the equilibrium. Michael addition therefore provides a mechanism for the equilibration of *Z* alkenes to *E* alkenes.

Double bonds in rings

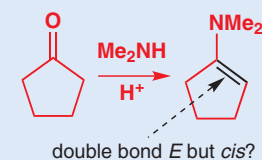
The smallest stable ring that can contain a *trans* double bond is cyclooctene—*trans*-cycloheptene can exist but is very unstable.



➔ You met ozone as a reagent for the oxidative cleavage of $\text{C}=\text{C}$ double bonds in Chapter 19. The products have carbonyl groups at the ends of the old alkene. The mechanism is described in more detail in Chapter 34.

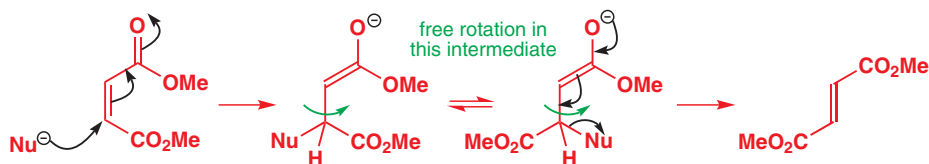
Nomenclature alert

Beware! The terms *cis* and *trans* do not always translate directly into *Z* and *E*. Consider the preparation of an enamine from cyclohexanone, which forms a double bond that you'd probably call *cis* (it's in a ring). But applying the rigorous rules laid down for *E/Z* nomenclature (p. 405), it is *E*. The same is true for the green double bond in the Birch reduction product above. As with the useful terms *syn* and *anti* (Chapter 14), there are no rigid rules for deciding whether a double bond is *cis* or *trans*. So there must be a diagram to make things clear if you use *cis* and *trans*.



■ This reaction is, of course, another simple example of the type we have just been discussing: the *Z* alkene arises from the cyclic starting material.

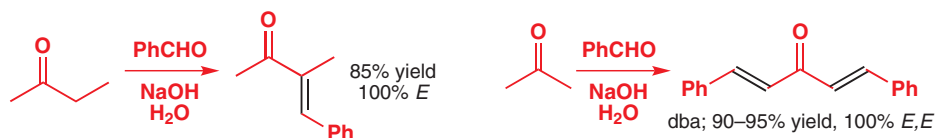
■ For this reason, it can be very difficult to make *Z* alkenes conjugated to reactive electrophilic groups such as aldehydes.



Similar mechanisms account for the double bond geometry obtained in aldol reactions followed by dehydration to give α,β -unsaturated carbonyl compounds. Any *Z* alkene that is formed is equilibrated to *E* by reversible Michael addition during the reaction. The two examples which follow illustrate how effective this method is.

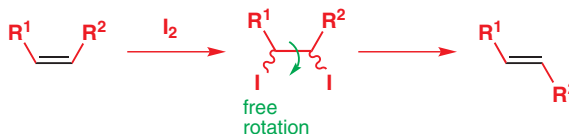
dba

The double aldol product from acetone and benzaldehyde, known as dibenzylidene acetone (dba), is a constituent of some sun-protection materials and is used in organometallic chemistry as a metal ligand.

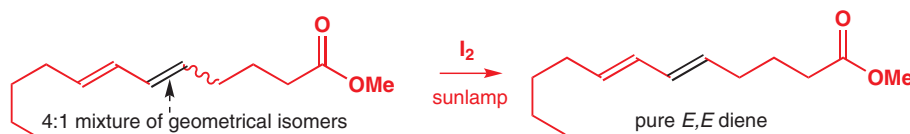


Equilibration of non-conjugated alkenes

Iodine will add reversibly not only to Michael acceptors but also to most other alkenes. It can therefore be a useful reagent for equilibrating double bond geometrical isomers.



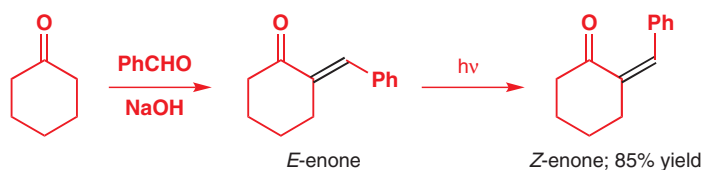
Some Japanese chemists needed the *E,E* diene below for a synthesis of a neurotoxic compound that they had isolated from poison dart frogs. Unfortunately, their synthesis (which used a Wittig reaction—described in detail later in this chapter) gave only 4:1 *E* selectivity at one of the double bonds. To produce pure *E,E* diene, they equilibrated the *E,Z* diene to *E,E* by treating with iodine and irradiating with a sun-lamp.



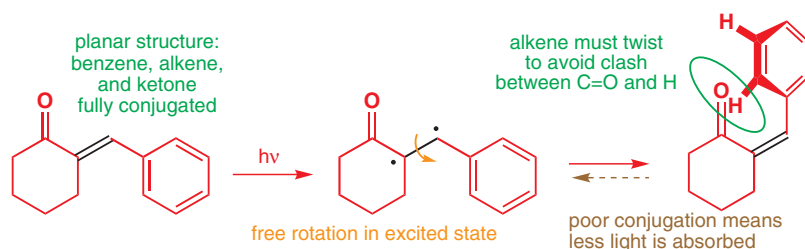
Using light to make *Z* alkenes from *E* alkenes

Light allows the interconversion of the two isomers of an alkene by promoting a π electron into the π^* orbital and transiently breaking the π bond, but the way light favours formation of the *Z* isomer is rather subtle. One difference between *cis* and *trans* alkenes is that the *trans* alkenes usually absorb light better than the *cis* alkenes—they absorb light of a higher wavelength and they absorb more of it, particularly when conjugated with carbonyl groups. Steric hindrance forces the *cis* alkene to twist about the σ bond joining the alkene to the carbonyl group and conjugation is then less efficient. In a mixture of *E* and *Z* alkenes, the *E* alkene is more prone to isomerization by light, so the *Z* isomer builds up in the mixture.

Here is an example. Aldol condensation of cyclohexanone and benzaldehyde gives pure *E* alkene for the reasons explained above. Irradiation with longer-wavelength UV light equilibrates this to the *Z* alkene in excellent yield.



It is not possible for the benzene ring and the enone system to be planar in the *Z*-enone and so they twist, making conjugation not as good as in the *E*-enone. Longer-wavelength light is absorbed only by the *E*-enone, which is continually equilibrated back to the excited state. Eventually, all the *E*-enone is converted to the *Z*-enone, which is not as efficiently excited by the light. The final mixture of *E*- and *Z*-enone is known as a 'photostationary state'.

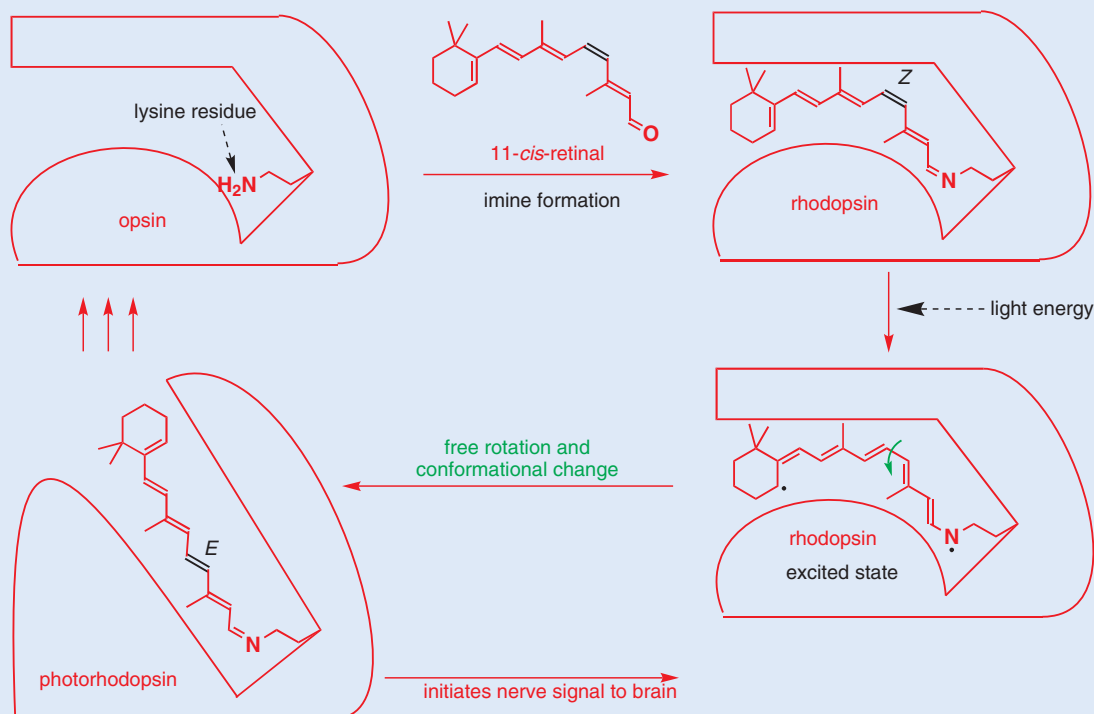


■ The excited state, in which an electron from the π orbital has been promoted to the π^* orbital, can be represented as a 'diradical'—the π bond is effectively broken, with the two electrons which formed it now residing, unpaired, on the two C atoms.

The chemistry of vision

The human eye uses a *cis* alkene, 11-*cis*-retinal, to detect light, and a *cis*–*trans* isomerism reaction is at the heart of the chemical mechanism by which we see. The light-sensitive pigment in the cells of the retina is an imine, formed by reaction of 11-*cis*-retinal with a lysine residue of a protein, opsin. Absorption of light by the opsin–retinal compound, known as rhodopsin, promotes one of

the electrons in the conjugated polyene system to an antibonding orbital. Free rotation in this excited state allows the *cis* double bond to isomerize to *trans*, and the conformational changes in the protein molecule that result trigger a cascade of reactions that ultimately leads to a nerve signal being sent to the brain.



E and Z alkenes can be made by stereoselective addition to alkynes

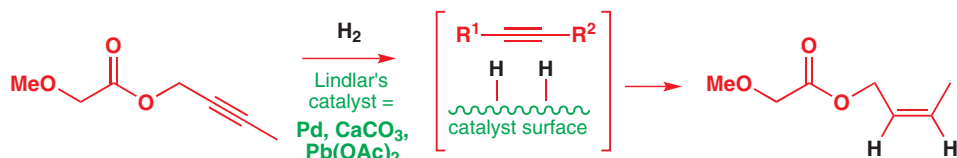
Alkenes can be made from alkynes by reduction or addition, and under the right conditions either the *Z* double bond or the *E* double bond can be formed stereoselectively.

Z-selective reduction of alkynes using Lindlar's catalyst

The *Z* alkene below was needed pure for studies on the mechanism of a rearrangement reaction. In Chapter 23 you met catalytic hydrogenation as a means of reducing alkenes to alkanes,

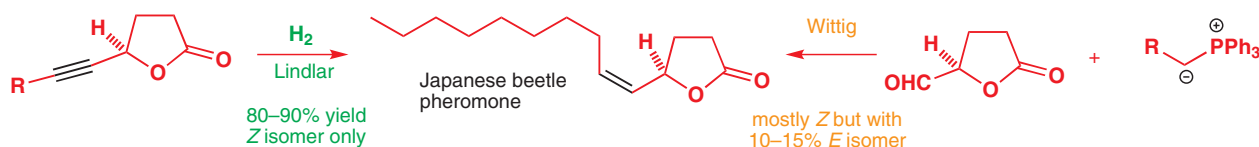
■ The reason that catalytic hydrogenation often results in *syn* addition of hydrogen to *alkenes* was discussed in Chapter 23.

and we introduced Lindlar's catalyst (palladium and lead acetate on a support of calcium carbonate) as a means of controlling chemoselectivity so that *alkynes* could be reduced to *alkenes*. What we did not emphasize then was that the two hydrogen atoms add to the alkyne in a *syn* fashion and the alkene produced is a *Z* alkene. The stereoselectivity arises because two hydrogen atoms, bound to the catalyst, are delivered simultaneously to the alkyne.



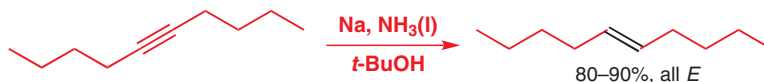
➔ We will discuss the selectivity of the Wittig reaction on p. 690.

The compound below is the pheromone of a destructive beetle. The synthetic pheromone can be used to trap the beetles, but it is active only as the *Z* isomer. Reduction of the alkyne with the Lindlar catalyst gives pure *Z* isomer, while the alternative way of making *Z* alkenes, the Wittig reaction, gives significant amounts of the *E* isomer.

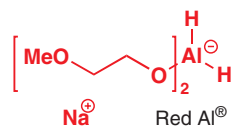
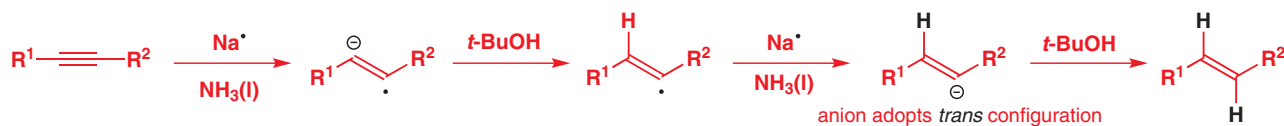


E-selective reduction of alkynes using sodium in liquid ammonia

The best way of ensuring *anti* addition of hydrogen across any triple bond is to treat the alkyne with sodium in liquid ammonia.



The sodium donates an electron to the LUMO of the triple bond (one of the two orthogonal π^* orbitals). The resulting radical anion can pick up a proton from the ammonia solution to give a vinyl radical. A second electron, supplied again by the sodium, gives an anion that can adopt the more stable *trans* geometry. A final proton quench by a second molecule of ammonia or by an added proton source (*t*-butanol is often used, as in the Birch reduction) forms the *E* alkene.

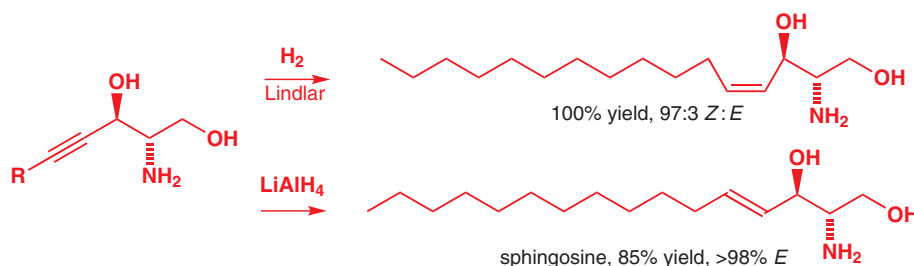


An alternative, and more widely used, method is to reduce alkynes with LiAlH₄ or the related reducing agent known as RedAl. This reaction works only if there is a hydroxy or an ether functional group near to the alkyne because it relies on delivery of the reducing agent to the triple bond through complexation to this oxygen atom.

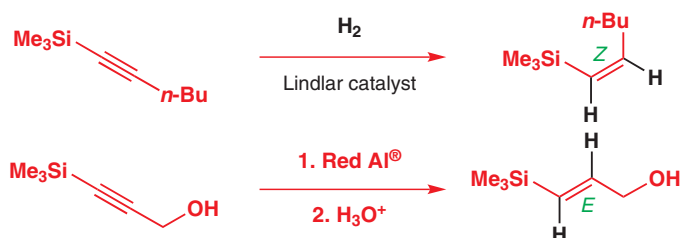


Making alkenes by addition to alkynes offers two distinct advantages. Firstly, the starting materials can often be made straightforwardly by alkylation of alkynyl anions. Secondly, the same alkyne can be used to make either *E* or *Z* alkene. In some early work on sphingosine (a constituent of cell membranes), some Swiss chemists needed to make both *E* and *Z*

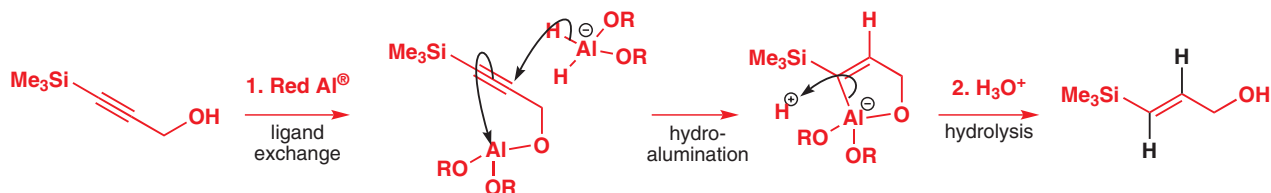
isomers of the naturally occurring compound. This was an easy task once they had made the alkyne.



Earlier in this chapter you were introduced to the significance of geometrically pure vinyl silanes, which can act as precursors to other alkenes. Controlled reduction of alkynyl silanes gives the corresponding vinyl silanes, with the method of reduction dictating the stereochemistry. Lindlar hydrogenation adds a molecule of hydrogen across the alkyne in a *cis* fashion to produce the *Z*-vinyl silane. RedAl reduction of a propargylic alcohol leads instead to the *E* isomer.



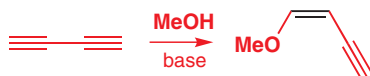
The mechanism of the aluminium hydride reductions with LiAlH_4 or RedAl involve a *trans* hydroalumination helped by coordination of Al to the triple bond and external nucleophilic attack. The regioselectivity of the hydroalumination is again determined by silicon: the electrophilic Al attacks the alkyne on the carbon bearing the silyl group (the *ipso* carbon).



■ In Chapter 40 you will meet further important ways of building alkenes by using Pd-catalysed coupling reactions of related vinylic compounds, in particular vinyl halides and vinyl stannanes. Many of the substrates for those reactions can be made using chemistry related to the reactions you see here.

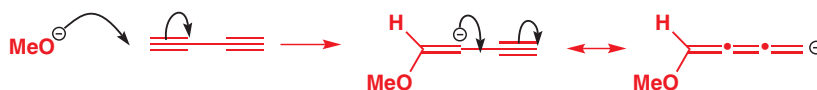
Addition of nucleophiles to alkynes

This rarer, and rather surprising, approach to *Z* alkenes can give excellent results, particularly in the addition of nucleophiles to butadiyne. The base-catalysed addition of methanol gives an excellent yield of *Z*-1-methoxybut-1-en-3-yne. This reaction is so easy to do that the product is available commercially. Notice that methanol adds once only: you would not expect nucleophiles to add to a simple alkyne and it is the conjugation that makes addition possible.

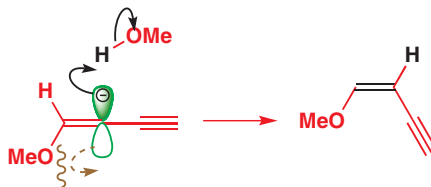


Methoxide ion adds to one of the alkynes to give a conjugated anion. The anion is linear with the negative charge delocalized into the second alkyne. The charge is therefore in a p orbital in the plane of the molecule, with a second conjugated π system oriented at right angles to the plane of the molecule.

■ We showed you in Chapter 19 (p. 435) how molecular orbitals explain why dienes are both more nucleophilic and more electrophilic than simple alkenes. The same arguments apply to diynes.



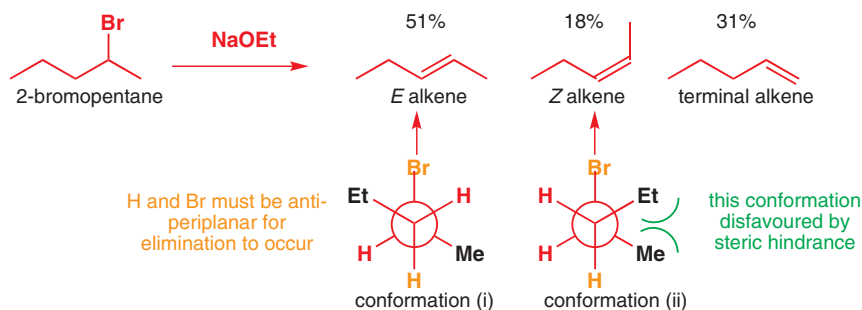
When the anion reacts with a molecule of methanol, protonation occurs on the lobe of the p orbital away from the MeO group and the *Z* alkene is formed.



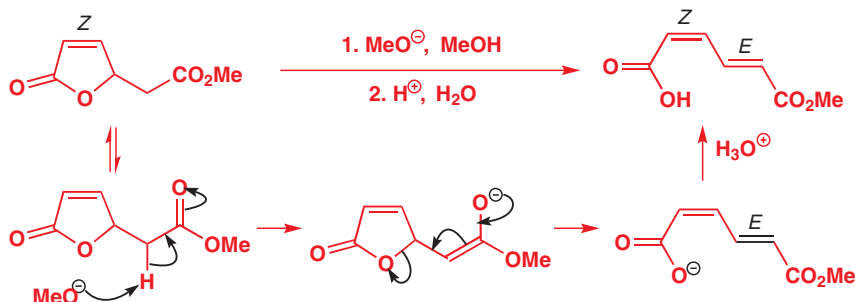
Predominantly *E* alkenes can be formed by stereoselective elimination reactions

➡ If you are unclear on the difference between *stereoselective* and *stereospecific* reactions, or between *kinetic* and *thermodynamic* control, go back and re-read Chapters 12 and 17—these concepts are very important for this chapter. The anti-periplanar transition state preferred by E2 reactions is described on p. 395.

In Chapter 17 you saw that E1 elimination reactions usually give mainly *E* alkenes (there's an example earlier in this chapter) because the transition state leading to an *E* double bond is lower in energy than that leading to a *Z* double bond. In other words, E1 reactions are **stereoselective**, and their stereoselectivity is **kinetically controlled**. E2 reactions are similar if there is a choice of protons that can be removed: the *E* alkene is preferred, but a mixture is still formed. Again, this is kinetic control.



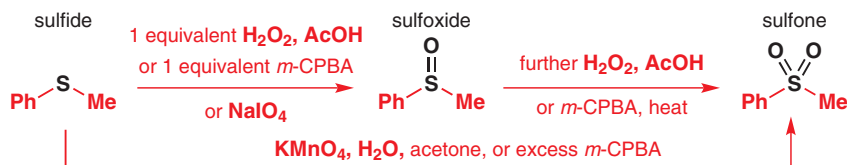
Both stereo- and regioselectivity are usually better in E1cB reactions, such as the opening of this unsaturated lactone in base. The double bond inside the ring remains *Z* but the new one, formed as the ring opens, prefers the *E* geometry. The transition state for the elimination step already looks like the product and prefers the *E* geometry for simple steric reasons.



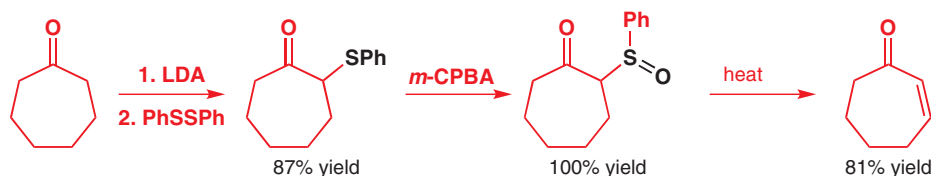
Sulfoxide elimination—oxidation to enones

Sulfoxides occupy a useful and interesting part of the middle ground between sulfides and sulfones—they are weakly nucleophilic, like sulfides (and can be alkylated with methyl

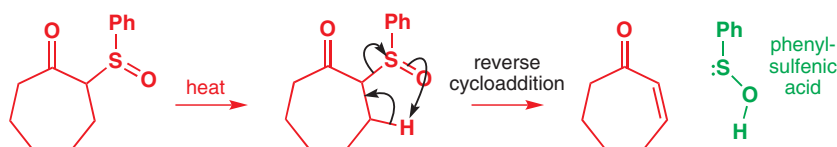
iodide to give sulfoxonium salts, as we saw on p. 667), but at the same time they stabilize anions almost as well as sulfones. They are easily made by controlled oxidation of sulfides and the chart below gives the main ways to get from sulfides to the two oxidized functional groups.



Sulfoxides can be used to make alkenes stereoselectively because sulfoxides next to electron-withdrawing or conjugating groups are unstable on heating, decomposing by an elimination process. The rather unstable phenylsulfenic acid (PhSOH) is eliminated and the reaction occurs partly because of the creation of conjugation and partly because PhSOH decomposes to volatile products. The starting material was made from cycloheptanone by sulfonylation of its enolate followed by oxidation.

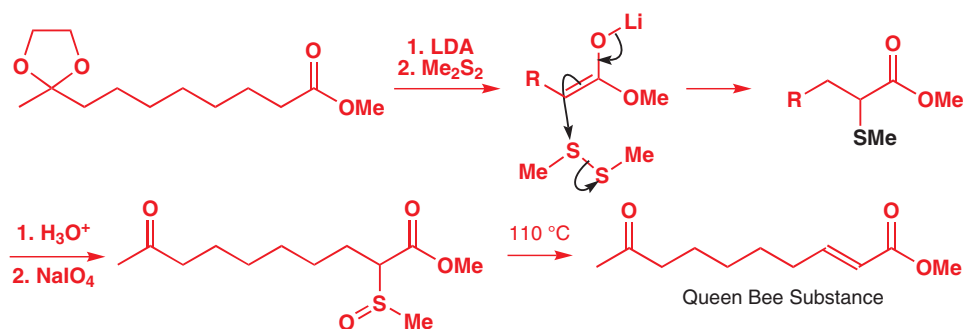


The elimination follows a type of mechanism we call a pericyclic reaction. Once you have read Chapter 34 you should be able to identify the reaction as a reverse cycloaddition, but for now you can just think of it as an elimination in which the proton is removed by the leaving group. The alkene product in this case is in a seven-membered ring—it has to be *cis*.



Interactive mechanism for sulfoxide elimination

This reaction provides a useful way of introducing a double bond next to a carbonyl group. Here it is in a synthesis of the Queen Bee Substance (the compound fed by the workers to those bee larvae destined to become queens). The compound is also a pheromone of the termite and is used to trap these destructive pests. The sulfur is introduced by reacting the enolate of the ester with the sulfur electrophile MeSSMe . Next, the protecting group is removed with acid, and the sulfide is oxidized to the sulfoxide with sodium periodate (NaIO_4) ready for elimination. Heating to 110°C then gives the Queen Bee Substance in 86% yield.



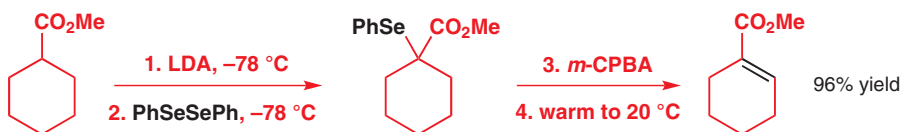
This elimination takes place more easily still when sulfur is replaced by a selenium— PhSe groups can be introduced by the same method, and oxidized to selenoxides with $m\text{-CPBA}$ at

■ Sulfur and selenium have many properties in common, and much sulfur chemistry is mirrored by selenium chemistry. In general, organoselenium compounds tend to be less stable and more reactive than organosulfur ones because the C–Se bond is even weaker than the C–S bond. They also have even fouler odours.

Marc Julia (1922–2010) was born in Paris, did his PhD at Imperial College, London, with Sir Derek Barton and then worked at the École Normale Supérieure in Paris.

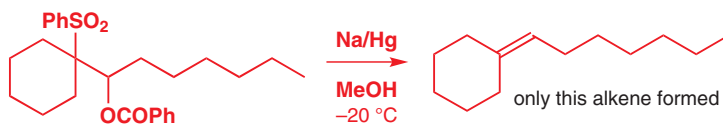
■ **Olefin** is an alternative name for alkene and **olefination** simply means alkene synthesis, usually by the formation of both σ and π bonds.

low temperature. The selenoxides are rarely isolated because the elimination takes place rapidly at room temperature.

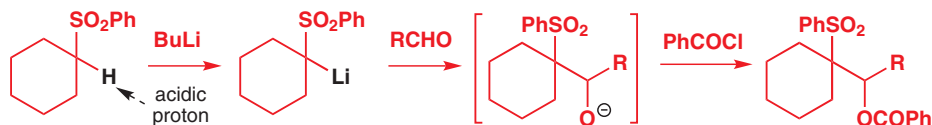


The Julia olefination is regiospecific and connective

Sulfoxide eliminations are a valuable way of introducing a double bond to an already intact carbon skeleton. The alkene synthesis we are about to show you is also based on sulfur chemistry, but is *connective*—the alkene is formed by joining together two separate fragments. It is called the **Julia olefination** and is probably the most important application of the sulfone-stabilized anions you saw earlier in the chapter. Only the alkene shown is formed, with the double bond joining the two carbons that carried the PhSO_2 and PhCO_2 groups. This elimination is promoted by a reducing agent, traditionally sodium amalgam (a solution of sodium metal in mercury) and works for a variety of compounds providing they have a phenylsulfonyl group adjacent to a leaving group.

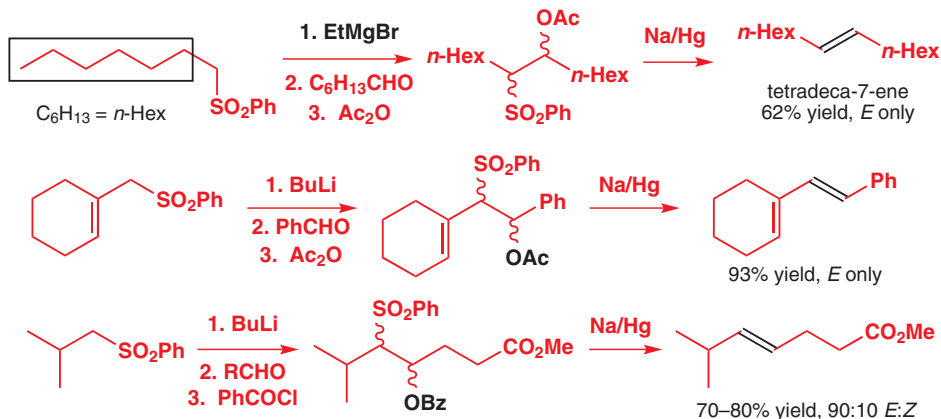


Common leaving groups are carboxylates such as acetate or benzoate, and the starting materials are very easily made. The sulfone-stabilized anion adds to aldehydes and a simple esterification step, which can be done in the same reaction vessel, introduces the acetate or benzoate group. This is how the starting material for the elimination above was made.



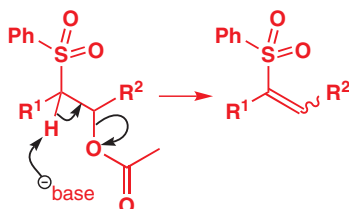
The Julia olefination is stereoselective

Here are the results of a few simple Julia olefinations. Notice that deprotonations can be with BuLi or EtMgBr and that the acylation step works with acetic anhydride or with benzoyl chloride. As you can see, they are all highly stereoselective for the *E* isomer, and the Julia olefination is one of the most important ways of making *E* double bonds connectively.

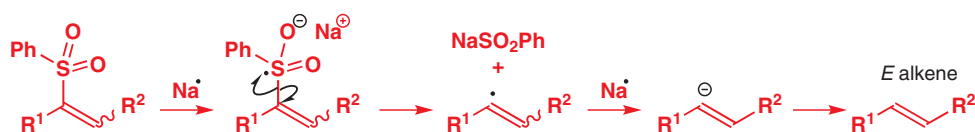


The reason for the *E* selectivity lies in the mechanism of the elimination. The details are not fully clear, but the first step, under the basic conditions of the reduction, appears to be the elimination of the acetate or benzoate ester to give a vinyl sulfone.

 Interactive mechanism for Julia olefination



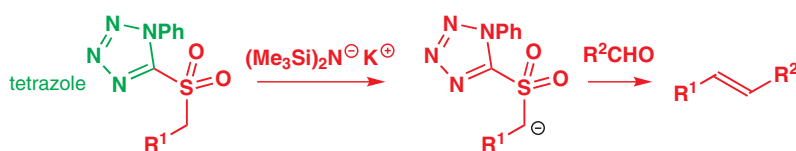
The stereochemistry of the vinyl sulfone does not matter because it is immediately reduced by an electron from sodium to give a vinyl radical. Much as you saw above, in the Birch reduction of alkynes, the vinyl radical collects a second electron and becomes a vinyl anion, which chooses to adopt the more stable *E* configuration before being protonated to give the predominantly *E* alkene.



We know that there must be an anion intermediate because the elimination is *not stereospecific*—in other words, whichever diastereoisomer of the starting material you use (all of the examples in this section have been mixtures of diastereoisomers) you always get the *E* alkene product.

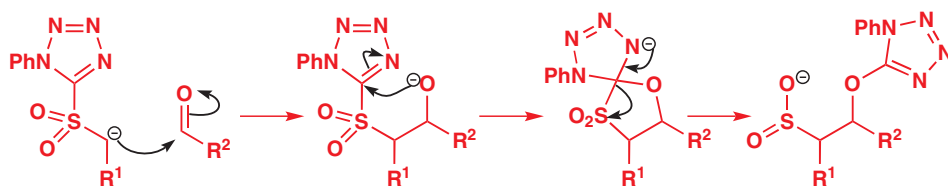
The one-step Julia olefination

The Julia reaction is remarkably versatile but it does need three steps to make the alkene: addition, acylation, and reduction. A more recent version of the reaction cuts this down to one by using not a phenylsulfone but instead a sulfone carrying an electron-deficient heterocycle, for example a tetrazole. The anion of the sulfone is made with a strong base (here potassium hexamethyldisilazide, KHMDS—see p. 635) and is added to an aldehyde to give an alkene directly.



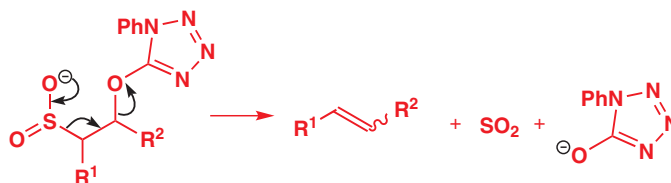
■ Tetrazoles look alarmingly unstable, but are in fact commonly used in medicinal chemistry. You will meet them again shortly, in the chapters on heterocyclic chemistry (29 and 30).

The elimination works because after the addition to the aldehyde, the alkoxide that is formed makes itself into a leaving group by grabbing the heterocyclic ring from the sulfur.



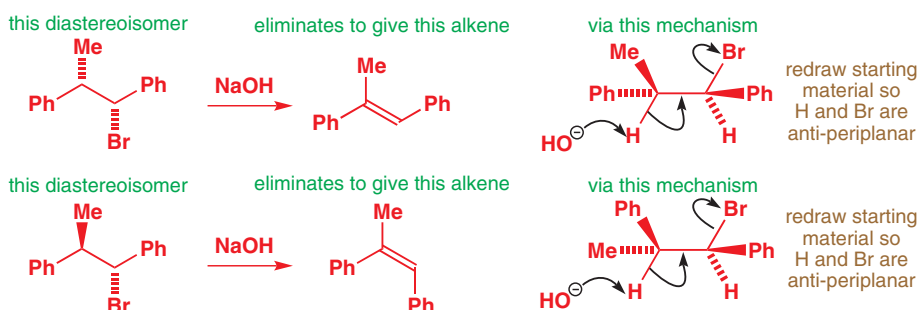
The final elimination is driven by loss of SO₂, and typically gives an *E* alkene, although by choosing carefully the base and solvent the selectivity can be tuned to give predominantly *Z*.

The one-step 'heterocyclic' modification of the Julia olefination was discovered by Marc Julia's brother, Sylvestre Julia, who also worked at the École Normale Supérieure. The use of tetrazolyl-sulfones was a contribution of Philip Kociński, and the reaction is sometimes known as the Julia–Kociński reaction.



Stereospecific eliminations can give pure single isomers of alkenes

You met a stereospecific elimination in Chapter 17. The requirement for the H and the Br to be anti-periplanar in the E2 transition state meant that the two diastereoisomers of this alkyl bromide eliminated to alkenes with different double bond geometries (p. 396).

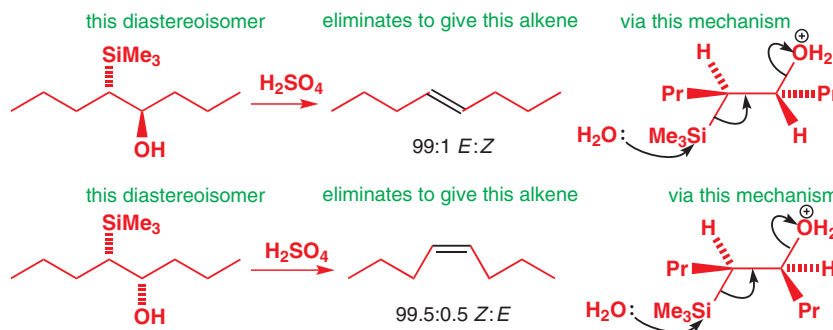


Interactive mechanism for stereospecific E2

However, reactions like this are of limited use—their success relies on the base's lack of choice of protons to attack. Logic dictates that only trisubstituted double bonds can be made stereospecifically in this way: the reaction must not have a choice of hydrogen atoms or an *E* alkene will result stereoselectively (as in the example on p. 684). The answer is, of course, to move away from eliminations involving H, and we can do this by returning to the Peterson elimination, which you met on p. 671.

The Peterson reaction is stereospecific

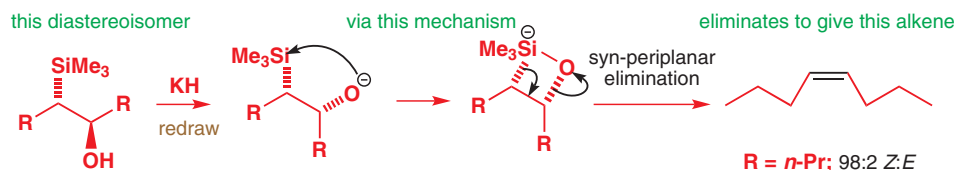
The stereospecificity of this elimination involving silicon arises because it is an E2 elimination proceeding via an anti-periplanar transition state. In principle, it can therefore be used to make single geometrical isomers of alkenes, the geometry depending on the relative stereochemistry of the starting material. However, this use of the Peterson reaction is limited by difficulties in making diastereoisomerically pure starting materials.



Interactive mechanism for stereospecific Peterson elimination

There is another, complementary, version of the Peterson reaction that uses base to promote the elimination. The starting materials are the same as for the acid-promoted Peterson reaction. When base (such as sodium hydride or potassium hydride) is added, the hydroxyl group is deprotonated and the oxyanion attacks the silicon atom *intramolecularly*. Elimination takes

place this time via a *syn-periplanar* transition state—it has to because the oxygen and the silicon are now bonded together, and it is the strength of this bond that drives the elimination forward.



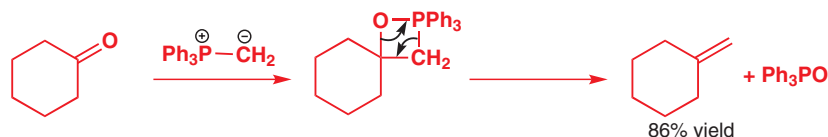
Interactive mechanism for stereospecific base-promoted Peterson elimination

The two versions of the Peterson reaction give opposite geometrical isomers from the same diastereoisomer of the starting material, so from any single diastereoisomer of hydroxy silane we can make either geometrical isomer of alkene product by choosing whether to use acid or base. The problem is still making those single diastereoisomers!

In Chapter 17 you saw that anti-periplanar transition states are usually preferred for elimination reactions because this alignment provides the best opportunity for good overlap between the orbitals involved. Syn-periplanar transition states can, however, also lead to elimination—and the base-promoted Peterson reaction should remind you of the Wittig reaction, which you first met in Chapter 11, with its four-membered cyclic intermediate. It is with the Wittig reaction, and a detailed discussion of its stereoselectivity, that we now finish this chapter.

Perhaps the most important way of making alkenes—the Wittig reaction

The Wittig reaction is another member of the class we have been talking about—it's an elimination that does not involve loss of H. You met it in Chapter 11, where we gave a brief outline of its mechanism.



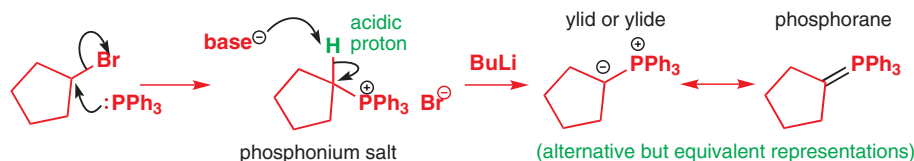
➔ This Wittig reaction appeared on p. 237. We also used the Wittig reaction to help us control enolates on p. 627 of Chapter 26.

Interactive mechanism for the Wittig reaction

Conceptually, the Wittig reaction is like the base-promoted Peterson reaction: it is a *syn* elimination, driven by the strength of an oxygen–heteroatom bond, although in this case the heteroatom is phosphorus. But the elimination step of the Wittig reaction occurs only from an intermediate and not from isolated starting materials. This intermediate is made *in situ* in the reaction and decomposes spontaneously: the Wittig reaction is therefore another connective alkene-forming reaction, but its simplicity makes it more widely used than the Julia or Peterson reactions.

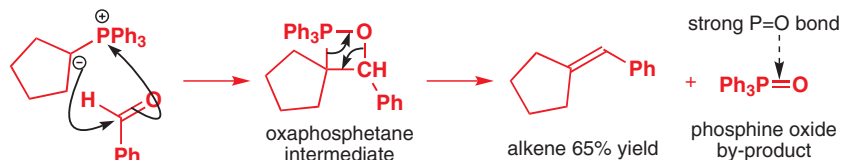
To understand the details of the reaction, we must start at the beginning. Phosphorus atoms, especially those that are positively charged or that carry electronegative substituents, can increase the acidity of protons adjacent to them on the carbon skeleton. Phosphonium salts (made in a manner analogous to the formation of ammonium salts from amines, in other words by reaction of an alkyl halide with a phosphine) can therefore be deprotonated by a moderately strong base to give a species known as an **ylid** (sometimes spelled **ylide**), carrying (formally) a positive and a negative charge on adjacent atoms. Ylids can alternatively be represented as doubly bonded species, called **phosphoranes**.

■ Phosphorus is like sulfur in this regard: you can compare the phosphonium ylid with the sulfonyl-stabilized anions you met earlier.



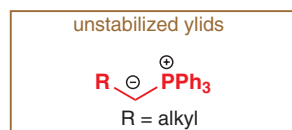
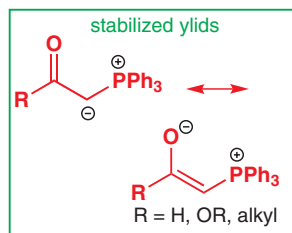
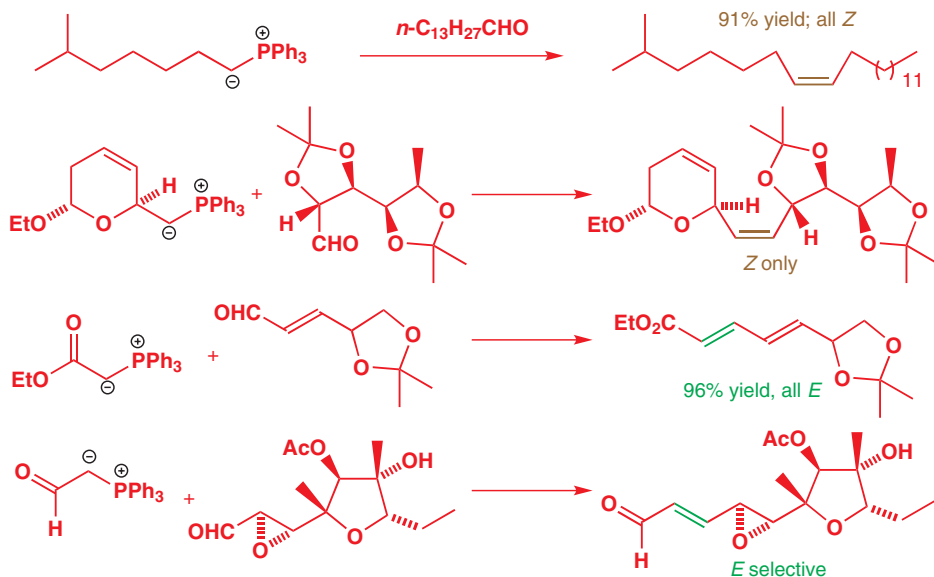
Ylids can be isolated, but are usually used in reactions immediately they are formed. They are nucleophilic species that will attack the carbonyl groups of aldehydes or ketones, generating the four-membered ring oxaphosphetane intermediates. Oxaphosphetanes are unstable: they undergo elimination to give an alkene (65% yield for this particular example) with a phosphine oxide as a by-product. The phosphorus–oxygen double bond is extremely strong and it is this that drives the whole reaction forward.

■ In Chapter 11, to help you understand the reaction, we showed the addition to the carbonyl group to make the four-membered ring taking place in two steps. These steps are probably in fact concerted (they both happen at the same time), and a better representation of the reaction is the single-step formation of the four-membered ring shown here.



Stereoselectivity in the Wittig reaction depends on the ylid

The Wittig reactions below were all used in the synthesis of natural products. You will notice that some reactions are *Z* selective and some are *E* selective. Look closer, and you see that the stereoselectivity is dependent on the *nature of the substituent* on the carbon atom of the ylid.



We can divide ylids into two types: those with conjugating or anion-stabilizing substituents adjacent to the negative charge (such as carbonyl groups) and those without. We call the first sort **stabilized ylids** because the negative charge is stabilized not only by the phosphorus atom but by the adjacent functional group—we can draw an alternative enolate-type structure to represent this extra stabilization. The rest we call **unstabilized ylids**.

• The stereochemistry of the Wittig reaction

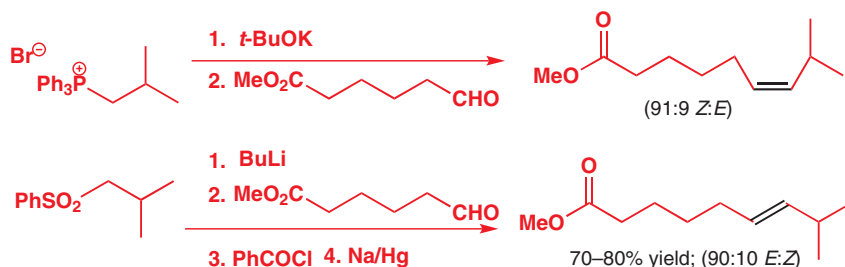
The general rule is:

- with *stabilized* ylids the Wittig reaction is *E* selective
- with *unstabilized* ylids the Wittig reaction is *Z* selective.

The *Z*-selective Wittig reaction

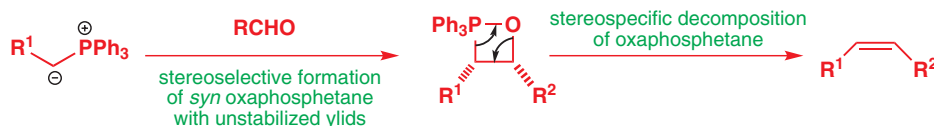
The *Z* selectivity observed with simple alkyl *R* groups is nicely complementary to the *E* selectivity observed in the Julia olefination. This complementarity was exploited by some chemists who wanted to make isomers of capsaicin (the compound that gives chilli peppers

their ‘hotness’) to follow up suggestions that capsaicin might be carcinogenic. The key intermediates in the synthesis of the *E* and *Z* isomers of capsaicin were the *E* and *Z* unsaturated esters shown below. By using a Wittig reaction with an unstabilized ylid it was possible to make the *Z* isomer selectively, whilst the Julia olefination gave the *E* isomer.

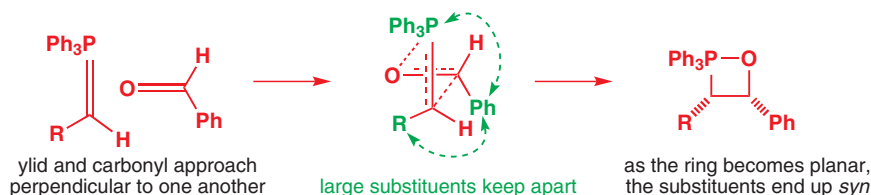


How can the *Z* selectivity in Wittig reactions of unstabilized ylids be explained? We have a more complex situation in this reaction than we had for the other eliminations we considered because we have two separate processes to consider: formation of the oxaphosphetane and decomposition of the oxaphosphetane to the alkene. The elimination step is the easier one to explain—it is stereospecific, with the oxygen and phosphorus departing in a *syn*-periplanar transition state. Addition of the ylid to the aldehyde can, in principle, produce two diastereoisomers of the intermediate oxaphosphetane. Provided that this step is irreversible, then the stereospecificity of the elimination step means that the ratio of the final alkene geometrical isomers will reflect the stereoselectivity of this addition step.

When R is not conjugating or anion-stabilizing, the *syn* diastereoisomer of the oxaphosphetane is formed preferentially, and the predominantly *Z* alkene that results reflects this. The *Z*-selective Wittig reaction therefore consists of a stereoselective first step, to form the *syn* oxaphosphetane, followed by a stereospecific elimination from this intermediate to give a *Z* alkene.



Why the *syn* oxaphosphetane is favoured with unstabilized ylids is the subject of much debate because the mechanism by which the oxaphosphetane is formed is not entirely understood. One possible explanation relies on rules of orbital symmetry, which you will meet in Chapters 34 and 35—we need not explain them in detail here but there is good reason to believe that, if the ylid and carbonyl compound react together to give the oxaphosphetane in one step, they will do so by approaching one another at right angles. Here we have drawn the ylid in its phosphorane form, adding to benzaldehyde. Keeping the large substituents apart produces a transition state like that shown below, which (correctly) predicts that the oxaphosphetane will have *syn* stereochemistry.

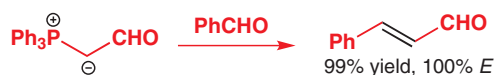


Interactive mechanism of *Z*-selective Wittig reaction

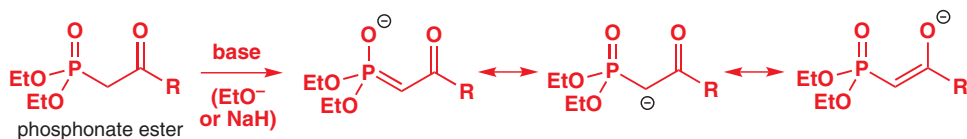
The *E*-selective Wittig reaction

Stabilized ylids, that is ylids whose anion is stabilized by further conjugation, usually with a carbonyl group, give *E* alkenes on reaction with aldehydes.

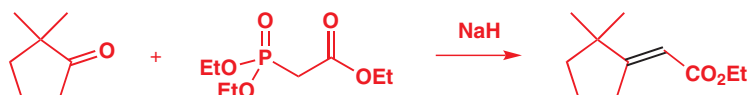
➔ These ylids are also enolates and were discussed in Chapter 26, p. 627.



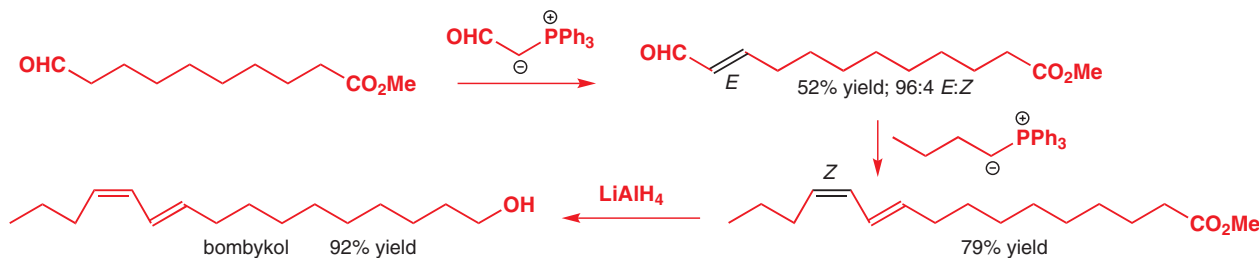
These stabilized ylids really are stable—this one, for example, can be recrystallized from water and the ylid is more stable than the phosphonium salt from which it might be made. This stability means though that they are not very reactive, and often it is better not to use the phosphonium salt but a phosphonate instead.



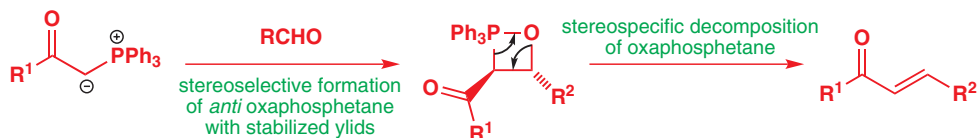
Phosphonate esters can be deprotonated with sodium hydride or alkoxide anions to give enolate-type anions that react well with aldehydes or ketones to give *E* alkenes. Alkene-forming reactions with phosphonates are called **Horner–Wadsworth–Emmons** (or Horner–Emmons, Wadsworth–Emmons, or even Horner–Wittig) **reactions**. This example is a reaction that was used by some Japanese chemists in the synthesis of polyzonimine, a natural insect repellent produced by millipedes.



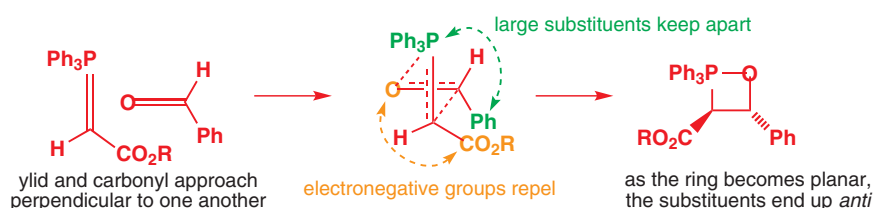
The synthesis below offers a nice illustration of the contrasting selectivity of the two classes of Wittig reagent. The female silkworm moth attracts mates by producing a pheromone known as bombykol. Bombykol is an *E,Z*-diene, and in this synthesis two successive Wittig reactions use first a stabilized and second an unstabilized ylid to control the stereochemistry of the product.



So why is there a change to *E* stereoselectivity when the ylid is stabilized? Again, the details are still unclear, and there are several possible explanations. Here we give one which is gaining ground, supported by recent experimental and computational evidence. It seems that, as with unstabilized ylids, the stereochemistry of the alkene product is determined by the stereochemistry of the intermediate oxaphosphetane, which with stabilized ylids must be *anti*.



It used to be thought that the formation of the *anti* oxaphosphetane was under thermodynamic control, but it now seems likely that it too is formed stereoselectively under kinetic control. The difference from the alkyl-substituted unstabilized ylids lies in the repulsion experienced between the polarized C=O bond of the aldehyde and the electronegative stabilizing group shown here as an ester CO₂R. As the four-membered ring flattens out, the CO₂R and Ph groups end up on opposite sides of the four-membered ring.



 Interactive mechanism for *E*-selective Wittig reaction

To conclude

In this chapter we have dealt for the first time with the problem of producing compounds as single stereoisomers—the stereoisomers concerned were geometrical isomers of alkenes. In future chapters we shall look in more detail at making stereoisomers, but we shall move out of two dimensions into three and consider reactions that exhibit diastereoselectivity and enantioselectivity. Methods for controlling stereochemistry in two and in three dimensions are closely related: single diastereoisomers are often made by addition reactions of single geometrical isomers of double bonds and, as you saw with the Peterson and Wittig reactions, single diastereoisomers can lead stereospecifically to single geometrical isomers.

● Summary of methods for making alkenes stereospecifically

To make <i>cis</i> (<i>Z</i>) alkenes	To make <i>trans</i> (<i>E</i>) alkenes
Wittig reaction of <i>unstabilized</i> ylid	Wittig reaction of <i>stabilized</i> ylid
Constrain the alkene in a ring	Equilibration to the more stable isomer
	Julia olefination
	Simple unselective elimination reactions
<i>Syn</i> addition of hydrogen across an alkyne	<i>Trans</i> selective reduction of alkyne
Peterson elimination	Peterson elimination

Further reading

If you want to read more about the elegant experiments that have been used to probe the structure of sulfonyl anions, see E. Block, *Reactions of organosulfur compounds*, Academic Press, New York, 1978.

Chemistry of Boron and Silicon is treated in S. E. Thomas, *Organic Synthesis: The Roles of Boron and Silicon*, Oxford Primer, OUP, Oxford, 1991.

For a recent discussion of the mechanism of the Wittig Reaction see R. Robiette, J. Richardson, V. K. Aggarwal, and J. N. Harvey, *J. Am. Chem. Soc.*, 2006, **128**, 2394.

For a comprehensive treatment of double bond geometry control see P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007 chapters 7–13 and the accompanying *Workbook*, also Wiley, 2008.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>